TECHNICAL SPECIFICATION · REVERSE-ENGINEERING REPORT

# RTT Compliance Datastore Update

Weekly referral-to-treatment (18-week) compliance per service: SystmOne download → RTT breach report → RTT Data workbook → ten compliance rows in the data store, reverse-engineered from an operator screen recording

| | |
|---|---|
| **Document** | Technical Specification — reverse-engineered legacy workflow (as-is) and target architecture (to-be) |
| **Subject workflow** | RTT compliance (Operational › RTT Compliance) for ten NHS services of Provide Wellbeing, updated weekly |
| **Primary evidence** | sixth/MEDDAD~1.MP4 — 1920×1080, 16 fps, 8:42 (523 s); camera off; recorded 14/08/2026 (clock 11:00–11:06 on screen). The short 8.3 file name means the original title was lost when the file was copied |
| **Secondary evidence** | MEDDAD~1.srt (matches this video) and MEDDAD~1.txt (TurboScribe) |
| **Timestamps** | Video positions; every figure matched to a verified frame at 0.25 s steps (fractions shown rounded down) |
| **Companion documents** | RTT_PreBuild_Client_Requirements.pdf (same folder); specs 01–05 |
| **Report date** | 16 October 2026 |
| **Status** | Draft for business-owner review. NHS numbers are blurred in every screenshot. **The operator changed a live formula during the recording**; see §4.3 |

**What the recording shows.** The operator explains the RTT Data workbook: one subject (Provide Wellbeing), ten services, a Power Query that loads the weekly RTT breach report (itself built from a SystmOne download), a patient-level RTT List and a Manage Compliance tab whose ten rows are pasted into the data store each week.

- **Rule:** each patient on the list has an RTT week `18-((Breach Date-TODAY())/7)`; a service's compliance is the share of its patients marked Compliant. RAG 10 if that share is above 92%, otherwise 2. Notes give the count of non-compliant patients.
- **Live edit to production logic (most important finding).** During the recording the operator deleted the *Breach Date 2* column ("we don't need a breach date"), which held service-specific adjustments (ECC Autism in Schools +70 days, Norfolk and Waveney ASD +42 days), producing #REF! errors. The RTT Week formula was then rewritten, first with ROUNDUP and then with **ROUNDDOWN**. With ROUNDDOWN the first patient, two days past their breach date, changed from **Not Compliant to Compliant**.
- **Thresholds disagree with the RAG formula:** the dashboard bands are 1 / 0.98 / 0.96 / 0.94 / 0.92, but `IF(N2>0.92,10,2)` marks exactly 92% as non-compliant, and the spoken rule is "less than 92 will be red".
- **Unexplained data:** "What's column 43? Can't actually remember"; weighting is always 1 ("Only Matt knows"); Every x months is 0.25 (weekly).
- **Usual leftovers:** Missing always TRUE, Unique Lookup with a trailing dash, TODAY()-based dates, hand-edited Last Updated (still 07/08/2026), "UNABLE TO REFRESH – linked workbook" in the data store.

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# 0. Evidence Base, Method & Reading Guide

## 0.1 Source artefacts

| Artefact | What it is | Used for |
|---|---|---|
| sixth/MEDDAD~1.MP4 | Teams recording, 8:42, Git-LFS object 39 MB (8.3 short file name) | Screenshots, step timings |
| sixth/MEDDAD~1.srt / .txt | Timed subtitles and transcript of the same meeting | Timestamps and wording |

## 0.2 Method and redaction

- Frames matched to verified grid frames at 0.25 s steps; formula strips read across every formula-bar change.
- **Redaction:** the NHS Number column of the RTT List and of the Power Query preview is blurred in every screenshot. Breach dates and services are kept (no direct identifier). Staff names appear in data-store screenshots, as in earlier specs.

# 1. Complete File, Workbook & Sheet Inventory

## 1.1 Files and systems

| # | File / system | Role | Evidence |
|---|---|---|---|
| S1 | **SystmOne** | Source of the RTT waiting-list download | Said 00:02:51–00:03:10 |
| F1 | **RTT breach report** (weekly; file name not shown) | Built by the operational team each week from the SystmOne download; analysed for breach prevention; source of the RTT List query | Said 00:02:26–00:03:19; Figure 9 [00:02:56] |
| F2 | **RTT Data** (.xlsx, User input folder) | Compliance tool: Manage Compliance, Requirements, Compliance Timeframes, RTT List (2) | Figure 2 [00:00:15] |
| F3 | **Compliance Data.xlsm** (Z:\Performance Dashboards and Data\Data Store) | Data store | Figure 23 [00:07:25], Figure 24 [00:07:59], Figure 25 [00:08:08] |

## 1.2 Tabs and query

| Object | Role |
|---|---|
| Manage Compliance | Output: ten compliance rows (one per service) with Compliance % and NC Patients helper columns |
| Requirements | Provide Wellbeing × 10 services |
| Compliance Timeframes | Catalogue, 0.25 months, thresholds |
| RTT List (2) / table RTT_List | Patient-level list loaded by the query, plus calculated RTT Week and Compliance |
| Query “RTT List” | Single query; last step Table.SelectRows(#"Filtered Rows", each [NHS Number] <> 0); preview “may be up to 7 days old” |

## 1.3 Overview screenshots

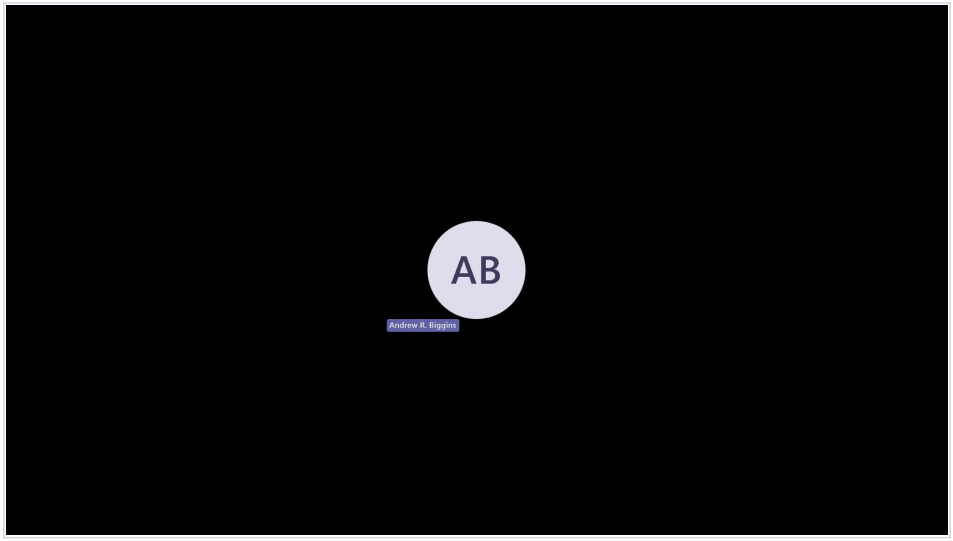



**Figure 1 [00:00:05] — Teams recording › (camera off):** Recording starts: “another video to show you how we update compliance data for RTT… for the whole of Provide Wellbeing”

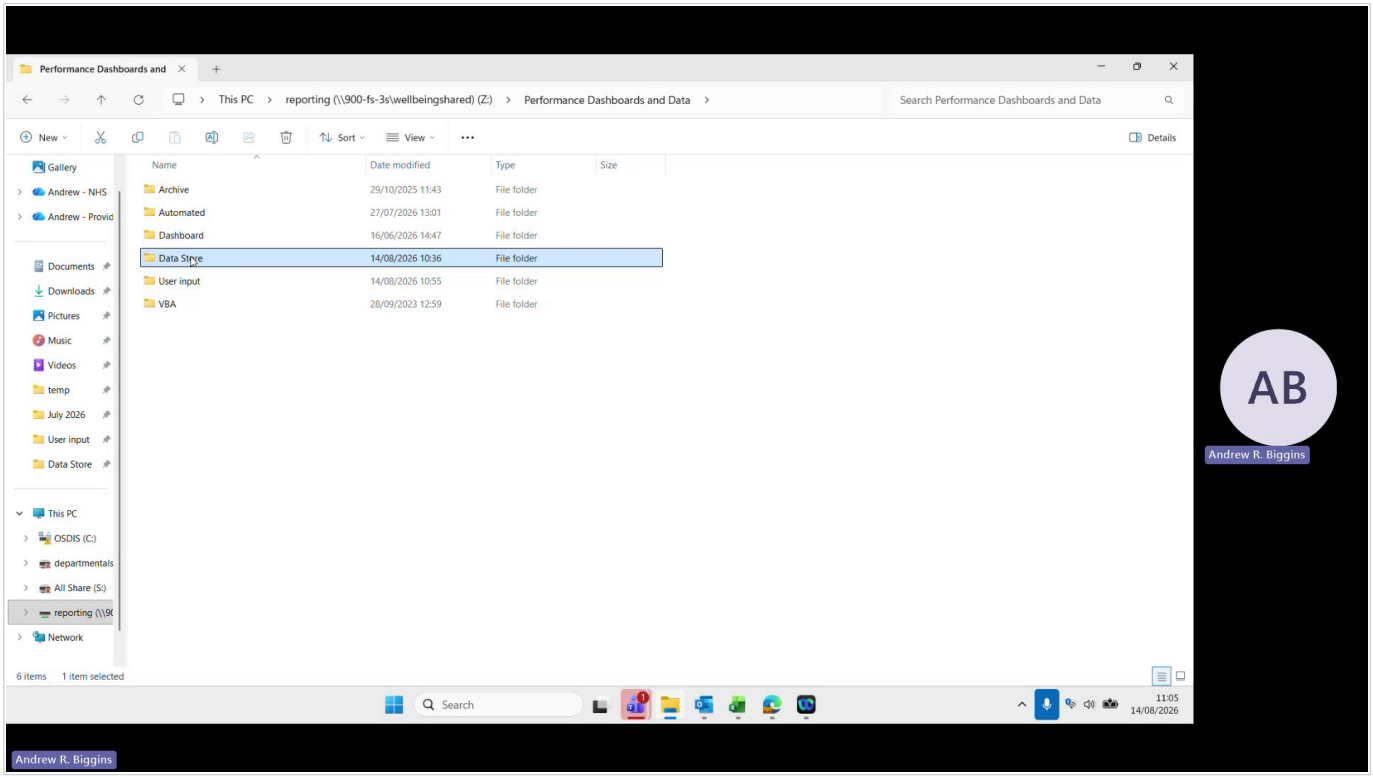



**Figure 23 [00:07:25] — File Explorer › Z:\Performance Dashboards and Data:** Reopening Compliance Data (“which I must have accidentally closed”): folders Archive, Automated, Dashboard, Data Store, User input, VBA

# 2. Exhaustive Column-Level Schema & Data Lineage Catalogue

## 2.1 Requirements and Compliance Timeframes

| Col | Header | Type | Origin | Lineage, rule and evidence |
|---|---|---|---|---|
| A–B | Name, Compliance Type | Text | MAN | Provide Wellbeing × 10 services (Figure 3 [00:01:14]) |
| C–E | Required, Core, Notes | | MAN | TRUE, Core, empty |
| F | Unique Lookup | Text | CALC | `=CONCAT([@Name],"-",[@[Compliance Type]])` |
| G | Active | Boolean | MAN | TRUE typed; “we can switch off services here” |
| TF B | Every x months | Decimal | REF | 0.25 (weekly) (Figure 4 [00:01:25]) |
| TF D–E | Compliance Group 1–2 | Text | REF | Operational › RTT Compliance |
| TF H | Unique Lookup | Text | CALC | Group1-Group2-Group3 → “Operational-RTT Compliance-” (Figure 5 [00:02:02]) |
| TF I | Missing | Boolean | CALC | Always TRUE (searches its own table) (Figure 6 [00:02:04]) |
| TF J–N | Threshold 1–5 | Decimal | REF | 1, 0.98, 0.96, 0.94, 0.92 (Figure 7 [00:02:10]) |

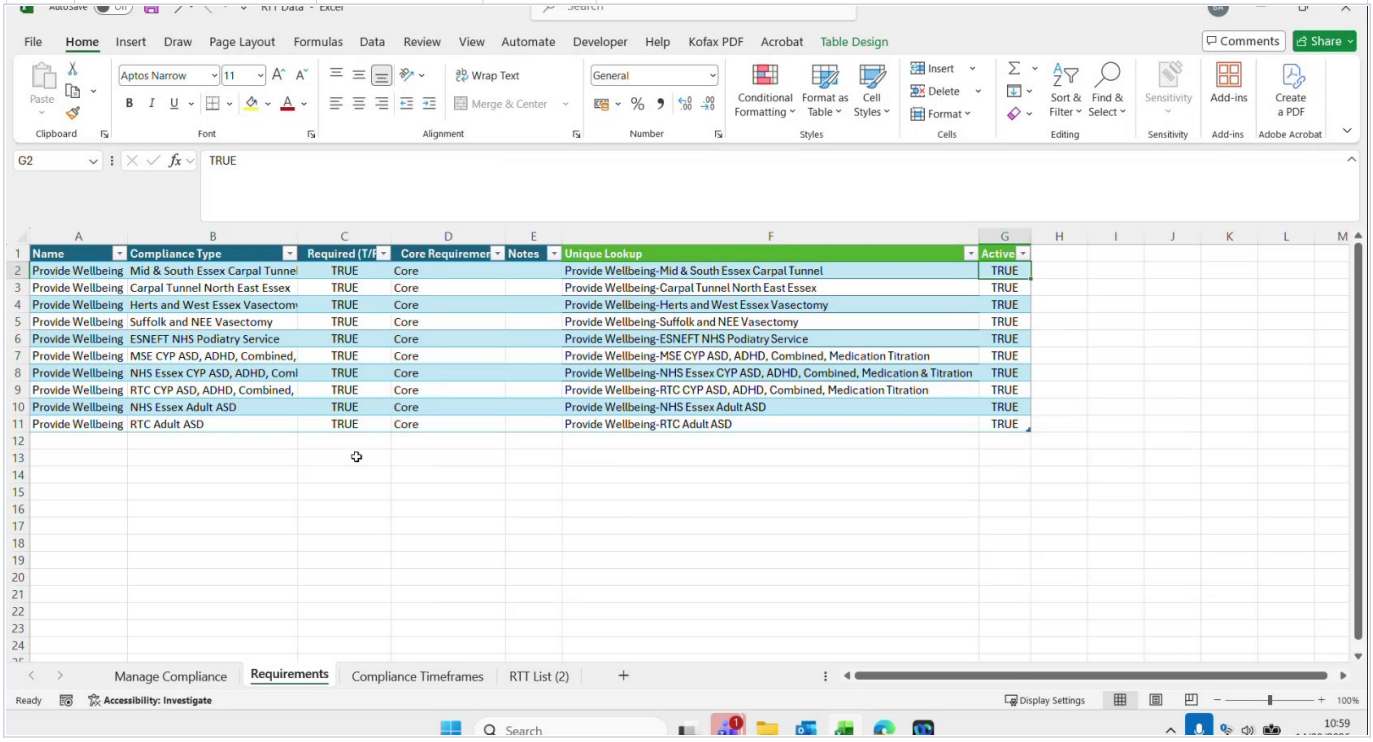



**Figure 3 [00:01:14] — RTT Data.xlsx › Requirements:** Requirements: Provide Wellbeing × 10 services (Mid & South Essex Carpal Tunnel, Carpal Tunnel North East Essex, Herts and West Essex Vasectomy, Suffolk and NEE Vasectomy, ESNEFT NHS Podiatry, MSE / NHS Essex / RTC CYP ASD-ADHD, NHS Essex Adult ASD, RTC Adult ASD); Required TRUE, Core, Unique Lookup =CONCAT([@Name],"-",[@[Compliance Type]]), Active TRUE (typed)

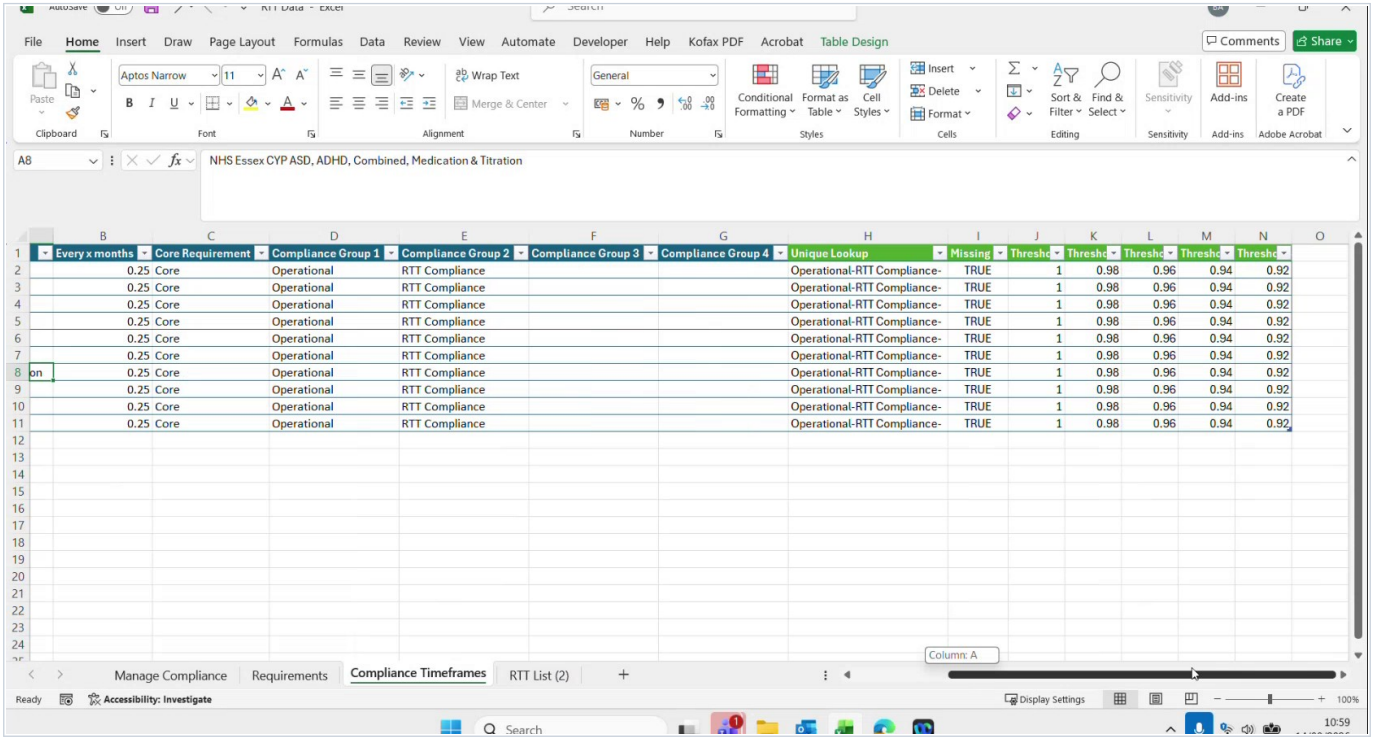



**Figure 4 [00:01:25] — RTT Data.xlsx › Compliance Timeframes:** Compliance Timeframes: Every x months = 0.25 (“we update this every week, so it’s four times a month”), Core, Operational › RTT Compliance

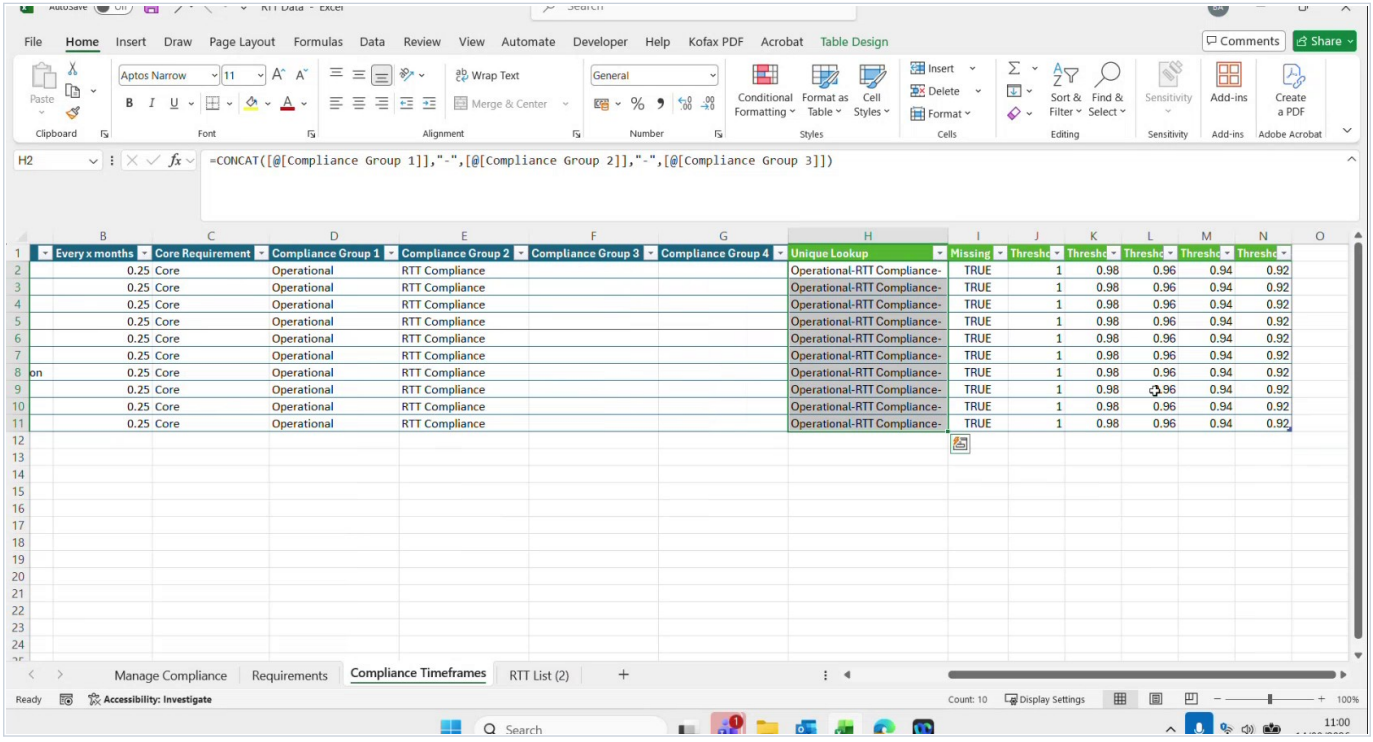



**Figure 5 [00:02:02] — RTT Data.xlsx › Compliance Timeframes:** Formula bar, H2 Unique Lookup =CONCAT([@[Compliance Group 1]],"-",[@[Compliance Group 2]],"-",[@[Compliance Group 3]]) → “Operational-RTT Compliance-” (“I just copied that from the previous sheet”)

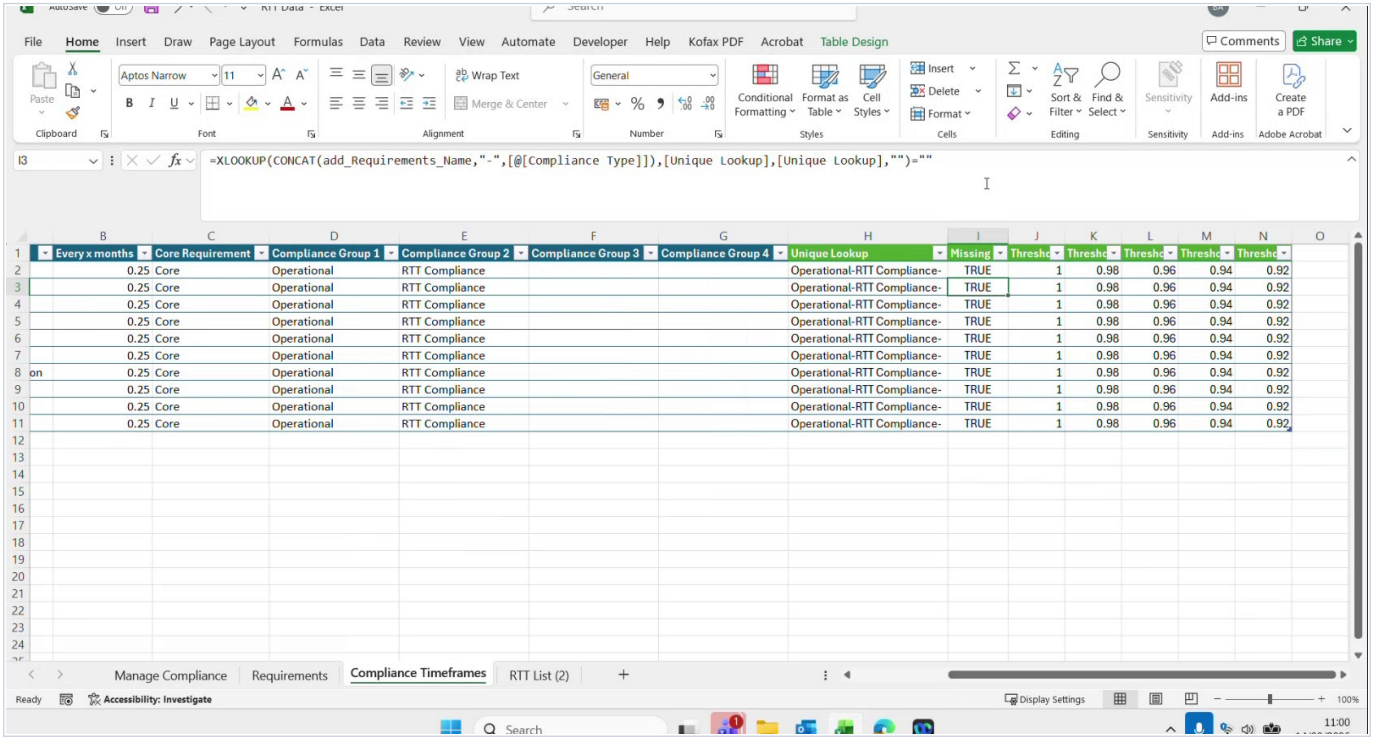



**Figure 6 [00:02:04] — RTT Data.xlsx › Compliance Timeframes:** Formula bar, I3 Missing =XLOOKUP(CONCAT(add_Requirements_Name,"-",[@[Compliance Type]]),[Unique Lookup],[Unique Lookup],"")="": searches its own table, so TRUE for every row (same defect as specs 02 and 05)

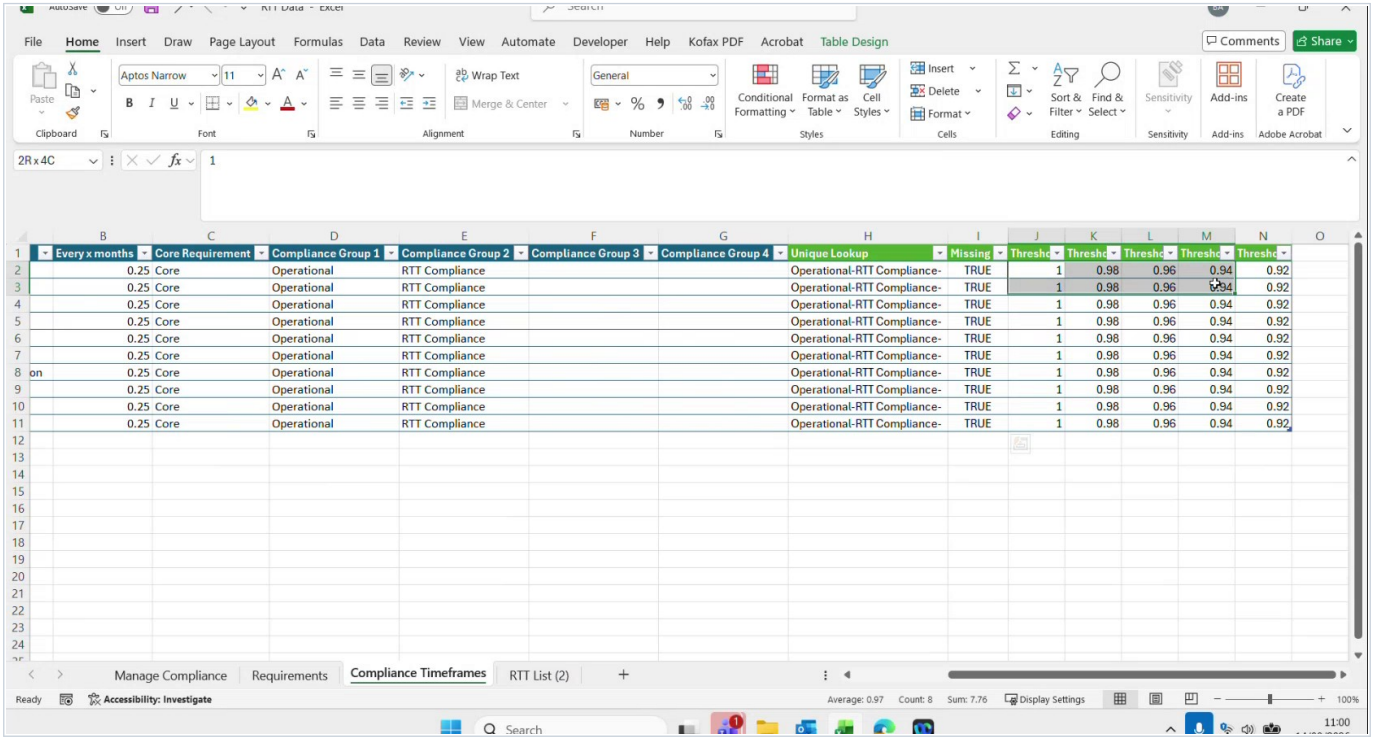



**Figure 7 [00:02:10] — RTT Data.xlsx › Compliance Timeframes:** Thresholds J:N = 1 / 0.98 / 0.96 / 0.94 / 0.92: green at 100%, red below 92% (“shades in between”)

## 2.2 RTT List (RTT_List table)

| Col | Header | Type | Origin | Lineage, rule and evidence |
|---|---|---|---|---|
| A | NHS Number | Text | PQ · **PID** | Patient identifier (blurred); rows with 0 removed by the query |
| B | Breach Date | Date | PQ | Date the 18-week limit is reached, from the breach report |
| C | Service | Text | PQ | Matches Requirements compliance types |
| D | Column43 | Decimal | PQ | e.g. 26.857, 11.857: “can’t actually remember” (Figure 10 [00:03:47]) |

| Col | Header | Type | Origin | Lineage, rule and evidence |
|---|---|---|---|---|
| E (before) | Breach Date 2 | Date | CALC | `=IF([@Service]="ECC Autism in Schools",[@[Breach Date]]+70,IF([@Service]="Norfolk and Waveney - ASD Assessment Service",[@[Breach Date]]+42,[@[Breach Date]]))` — deleted live (Figure 11 [00:04:01]) |
| F→E | RTT Week | Number | CALC | Before: `=18-(([@[Breach Date 2]]-TODAY())/7)`; after: `=ROUNDDOWN(18-(([@[Breach Date]]-TODAY())/7),0)` (Figure 12 [00:04:20], Figure 15 [00:05:41]) |
| G→F | Compliance | Text | CALC | Compliant / Not Compliant from RTT Week; formula not shown [Inferred: Not Compliant when RTT Week > 18] |

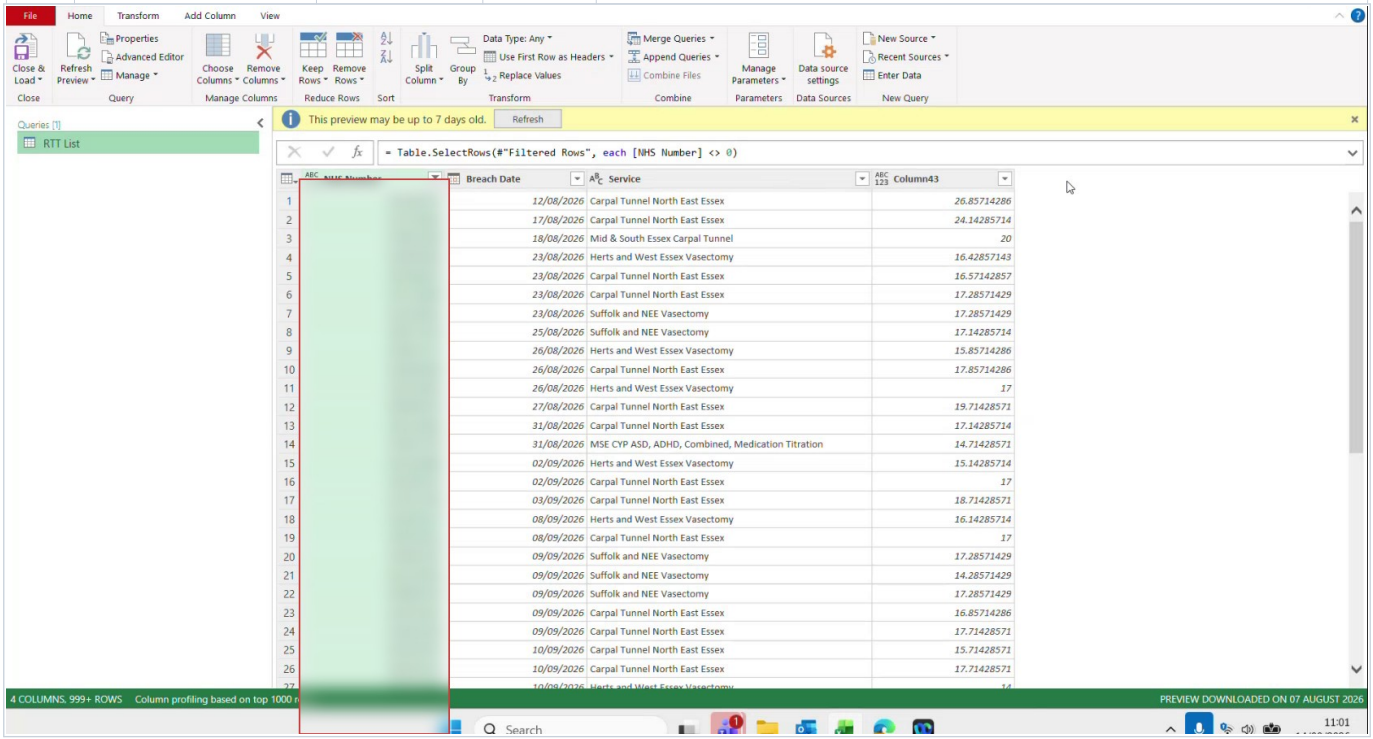



**Figure 9 [00:02:56] — RTT Data.xlsx › Power Query editor:** Query RTT List: last step =Table.SelectRows(#"Filtered Rows", each [NHS Number] <> 0); source is the weekly RTT breach report built from a SystmOne download (“This preview may be up to 7 days old”). NHS numbers blurred

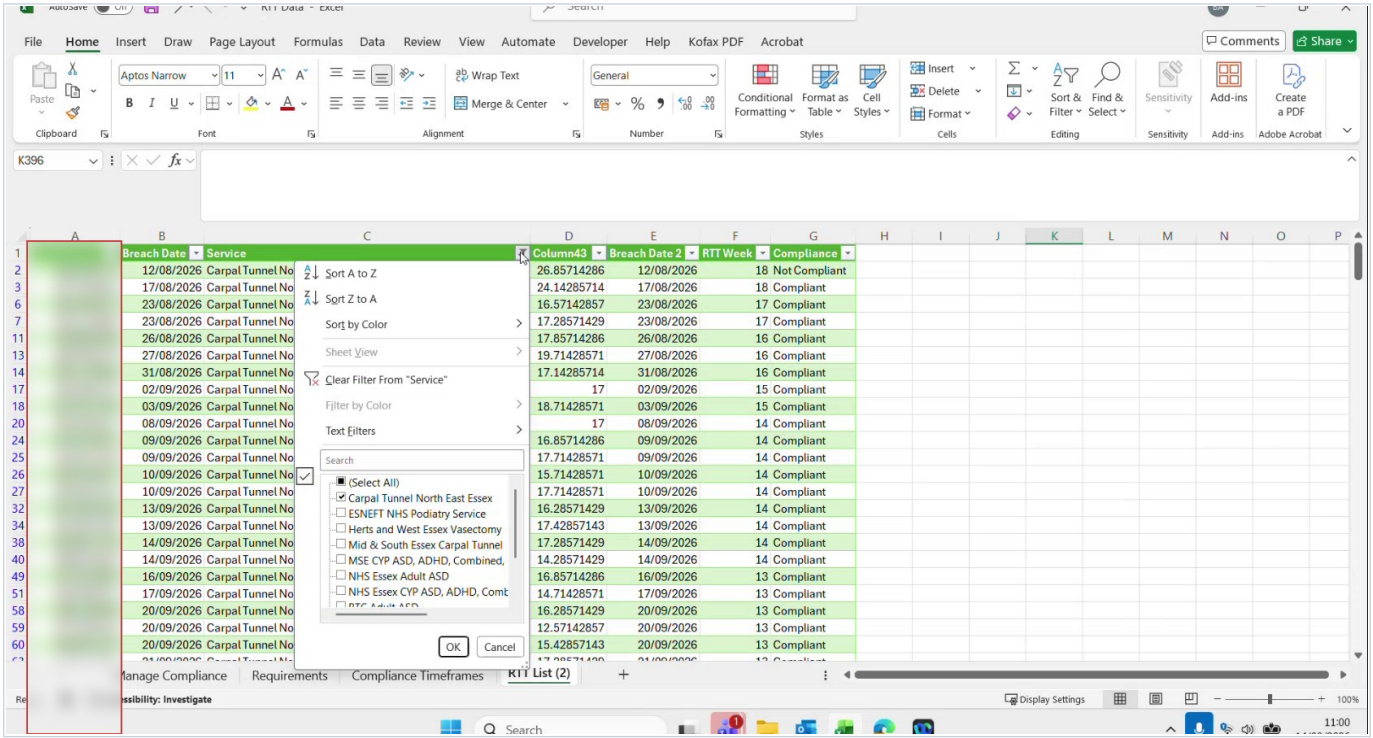



**Figure 8 [00:02:31] — RTT Data.xlsx › RTT List:** RTT List (query output RTT_List) with the Service filter menu open: NHS Number (blurred), Breach Date, Service, Column43, Breach Date 2, RTT Week, Compliance (Compliant / Not Compliant)

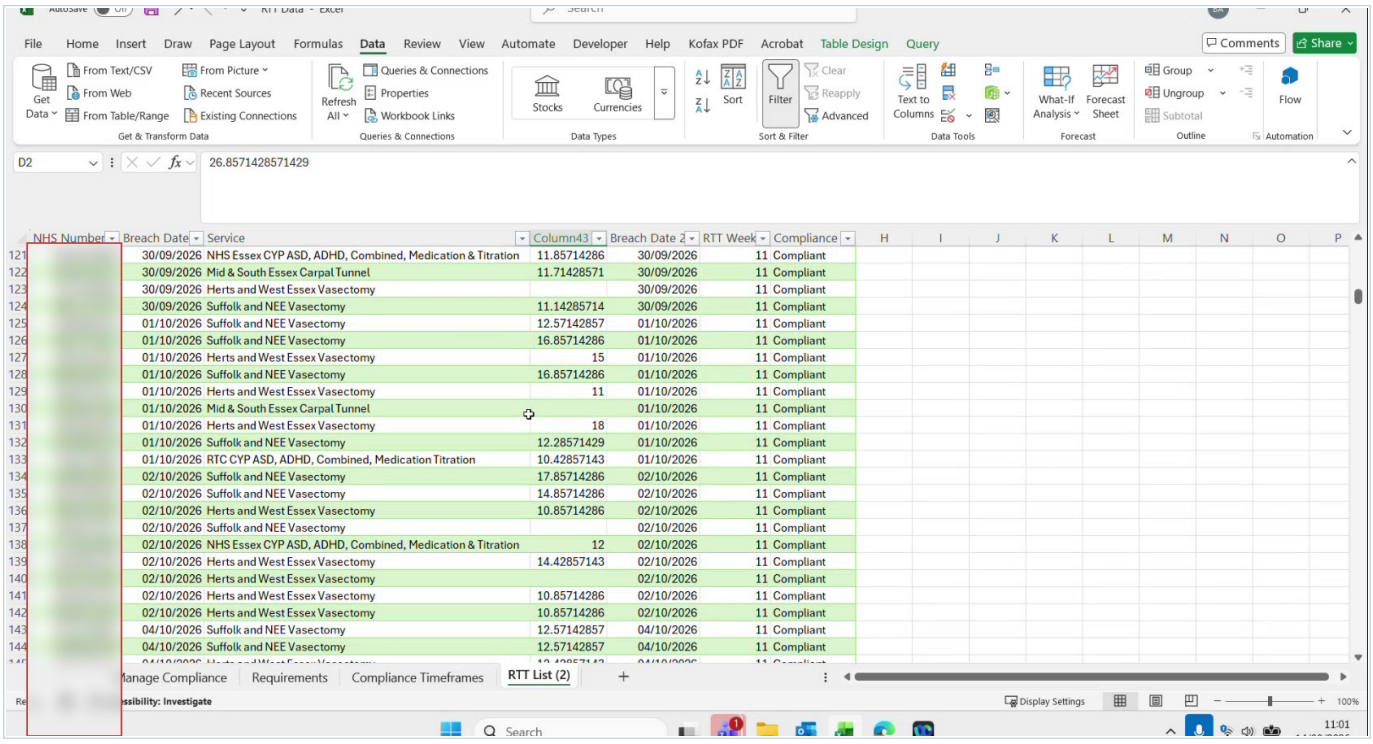



**Figure 10 [00:03:47] — RTT Data.xlsx › RTT List:** Column43 (e.g. 26.857, 11.857) — “What’s column 43? Can’t actually remember”: a weeks-waited value carried from the breach report [Inferred]

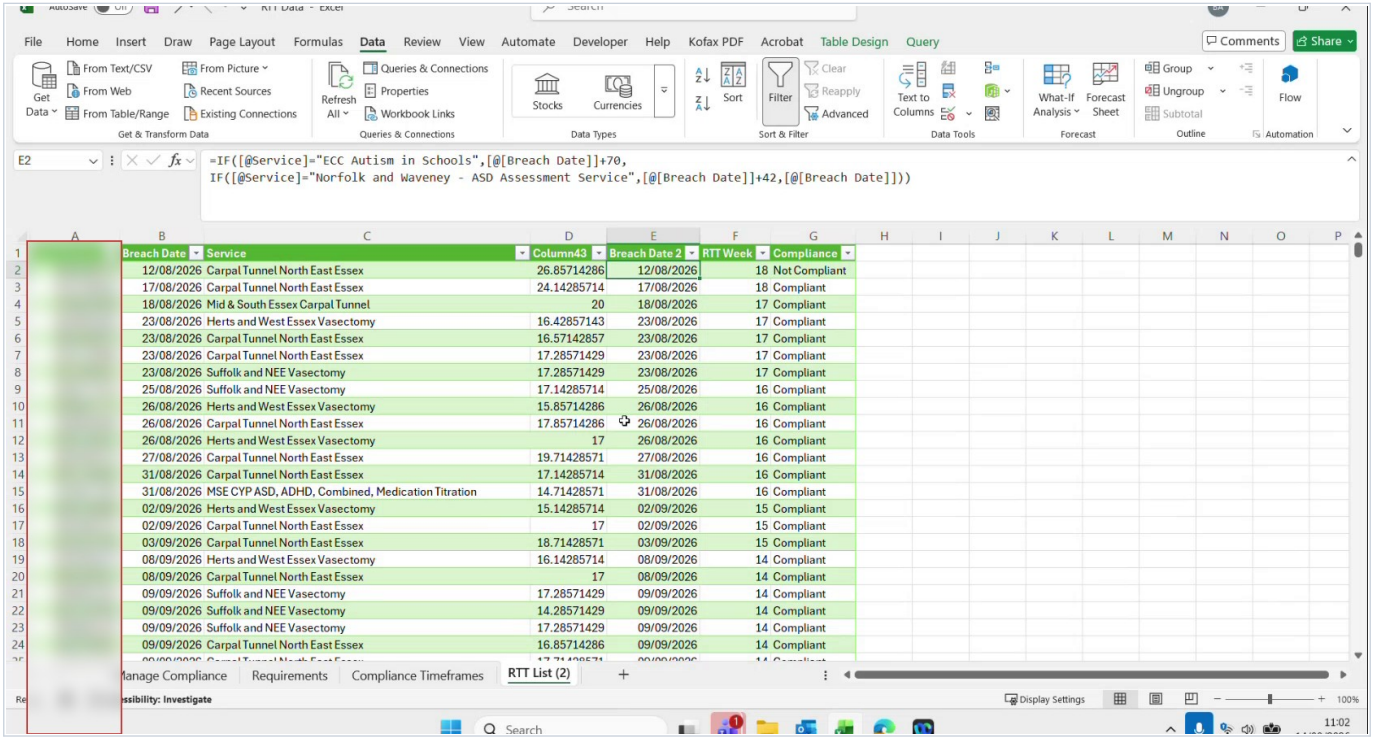



**Figure 11 [00:04:01] — RTT Data.xlsx › RTT List:** BEFORE the live edit: Breach Date 2 =IF([@Service]="ECC Autism in Schools",[@[Breach Date]]+70,IF([@Service]="Norfolk and Waveney - ASD Assessment Service",[@[Breach Date]]+42,[@[Breach Date]])) — service-specific breach-date adjustments

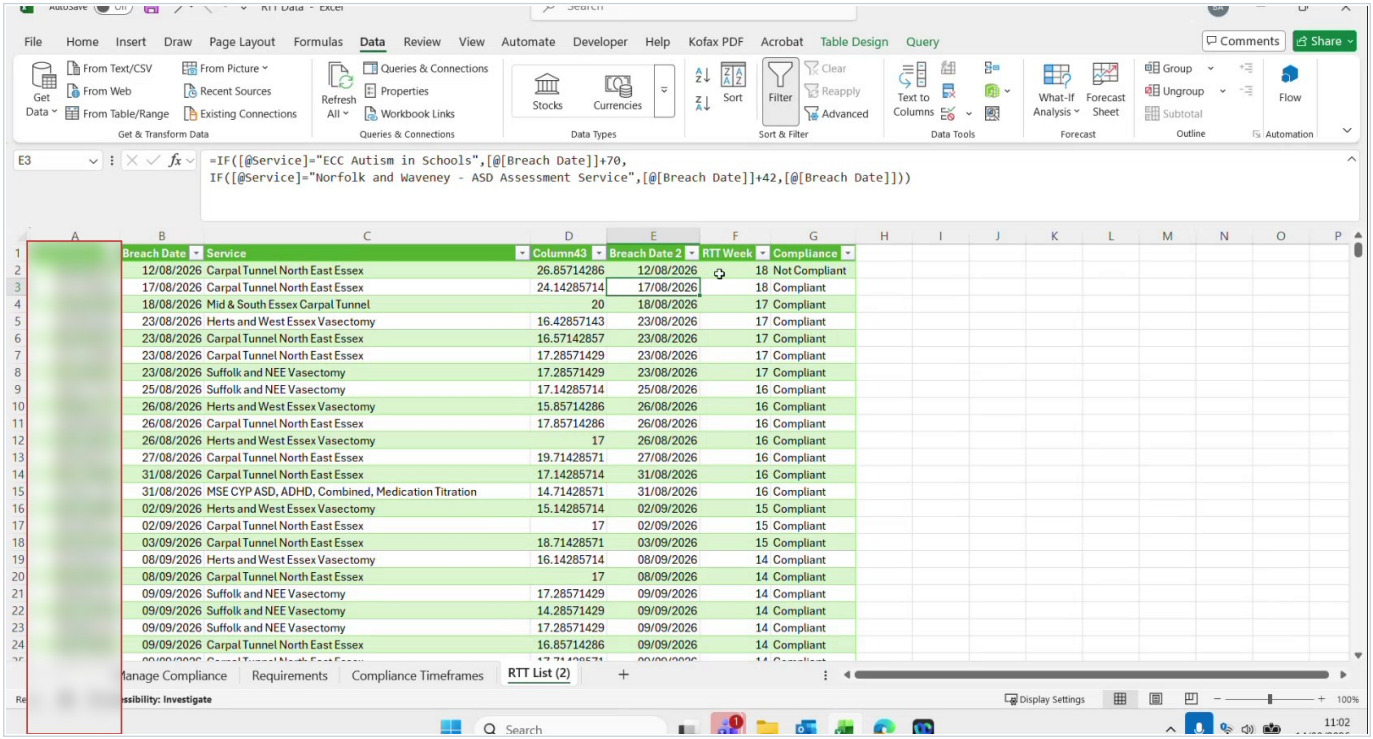



**Figure 12 [00:04:20] — RTT Data.xlsx › RTT List:** BEFORE: RTT Week F2 =18-(([@[Breach Date 2]]-TODAY())/7); first patient: breach 12/08/2026, RTT Week 18, Not Compliant

## 2.3 Manage Compliance (output rows)

| Col | Header | Type | Origin | Lineage, rule and evidence |
|---|---|---|---|---|
| A–B | Name, Compliance Type | Text | MAN | Provide Wellbeing; service |
| C | Completion Date | Date | CALC | `=TODAY()` (Figure 17 [00:05:52]) |
| D | Completion Status | Text | MAN | Completed (“all the time”) |
| E | RAG Rating | Number | CALC | `=IF(N2>0.92,10,2)` (Figure 18 [00:06:09]) |
| F | Notes | Text | CALC | `=IF(N2=1,"All Patients Currently Compliant",CONCAT(O2," Patient(s) Not Compliant"))` (Figure 20 [00:06:39]) |
| G | Managed By | Text | MAN | Provide Wellbeing |
| H | Due Date | Date | CALC | `=C2+7` (Figure 21 [00:06:48]) |
| I–L | Compliance Group 1–4 | Text | MAN | Operational, RTT Compliance |
| M | Weighting | Number | MAN | 1 (Figure 22 [00:07:13]) |
| N | Compliance | % | CALC | `=COUNTIFS(RTT_List[Service],B2,RTT_List[Compliance],"Compliant")/COUNTIF(RTT_List[Service],B2)` (Figure 16 [00:05:46]); #DIV/0! if a service has no patients |
| O | NC Patients | Integer | CALC | `=COUNTIFS(RTT_List[Service],B2,RTT_List[Compliance],"Not Compliant")` (Figure 19 [00:06:31]) |

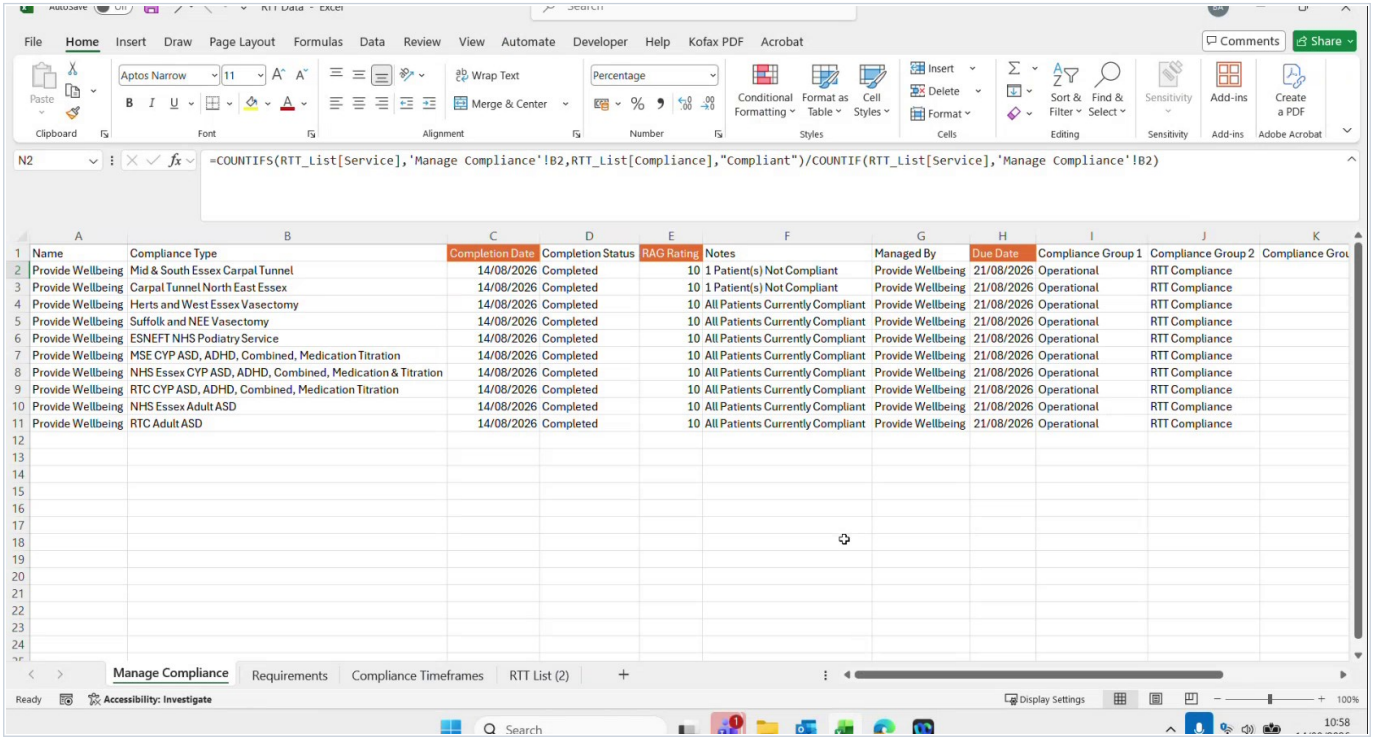



**Figure 2 [00:00:15] — RTT Data.xlsx › Manage Compliance:** Manage Compliance: one row per service (10), completed 14/08/2026, RAG 10, Notes such as “1 Patient(s) Not Compliant”, due 21/08/2026, Operational › RTT Compliance. Formula bar N2 (Compliance %) =COUNTIFS(RTT_List[Service],'Manage Compliance'!B2,RTT_List[Compliance],"Compliant")/COUNTIF(RTT_List[Service],'Manage Compliance'!B2)

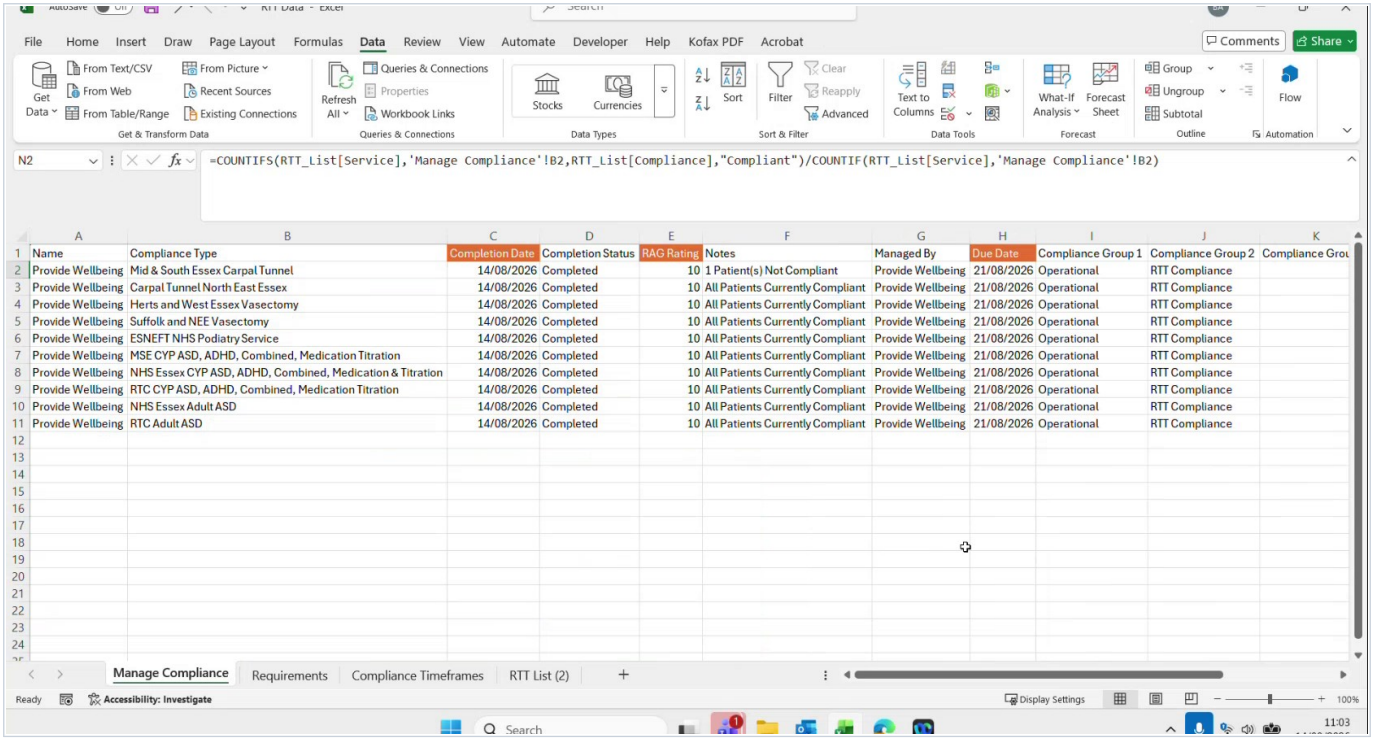



**Figure 16 [00:05:46] — RTT Data.xlsx › Manage Compliance:** Formula bar, N2 Compliance %: COUNTIFS(... "Compliant")/COUNTIF(...) per service from RTT_List

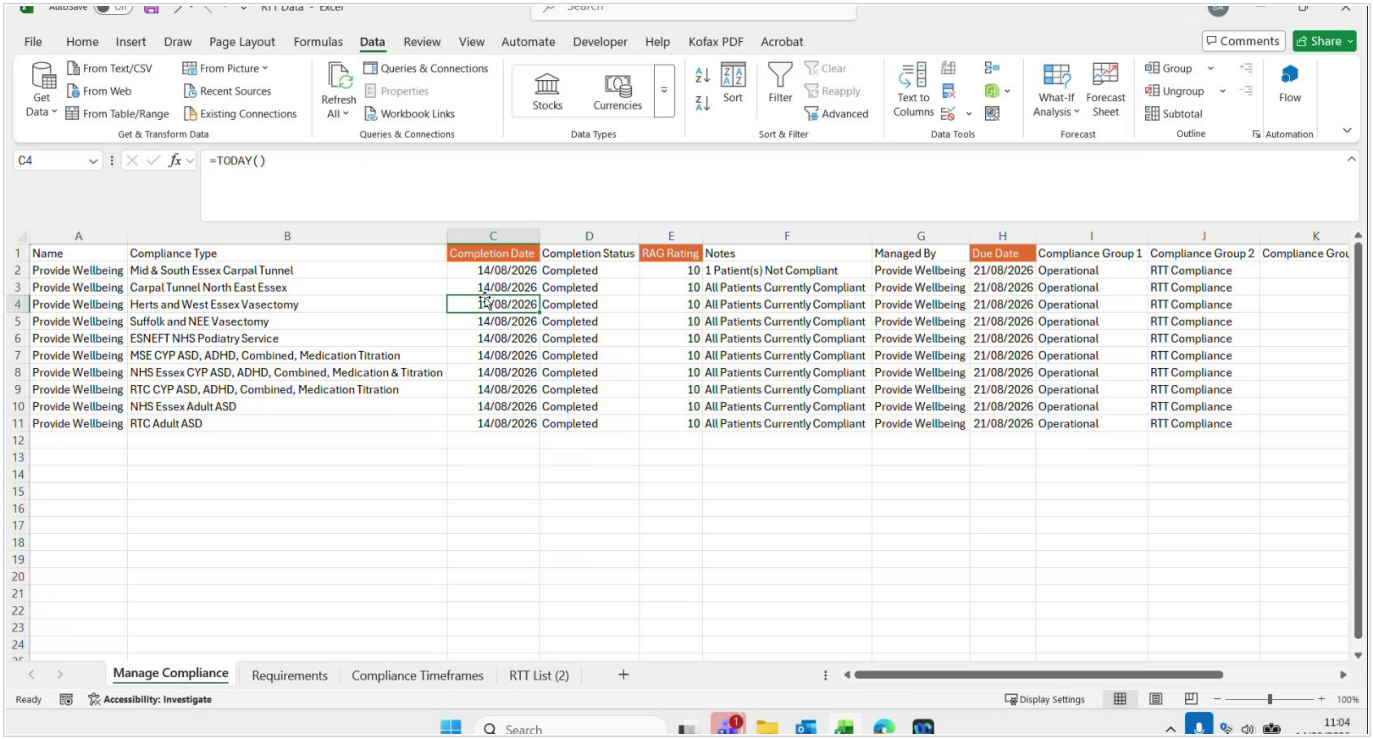



**Figure 17 [00:05:52] — RTT Data.xlsx › Manage Compliance:** Formula bar, C4 Completion Date =TODAY()

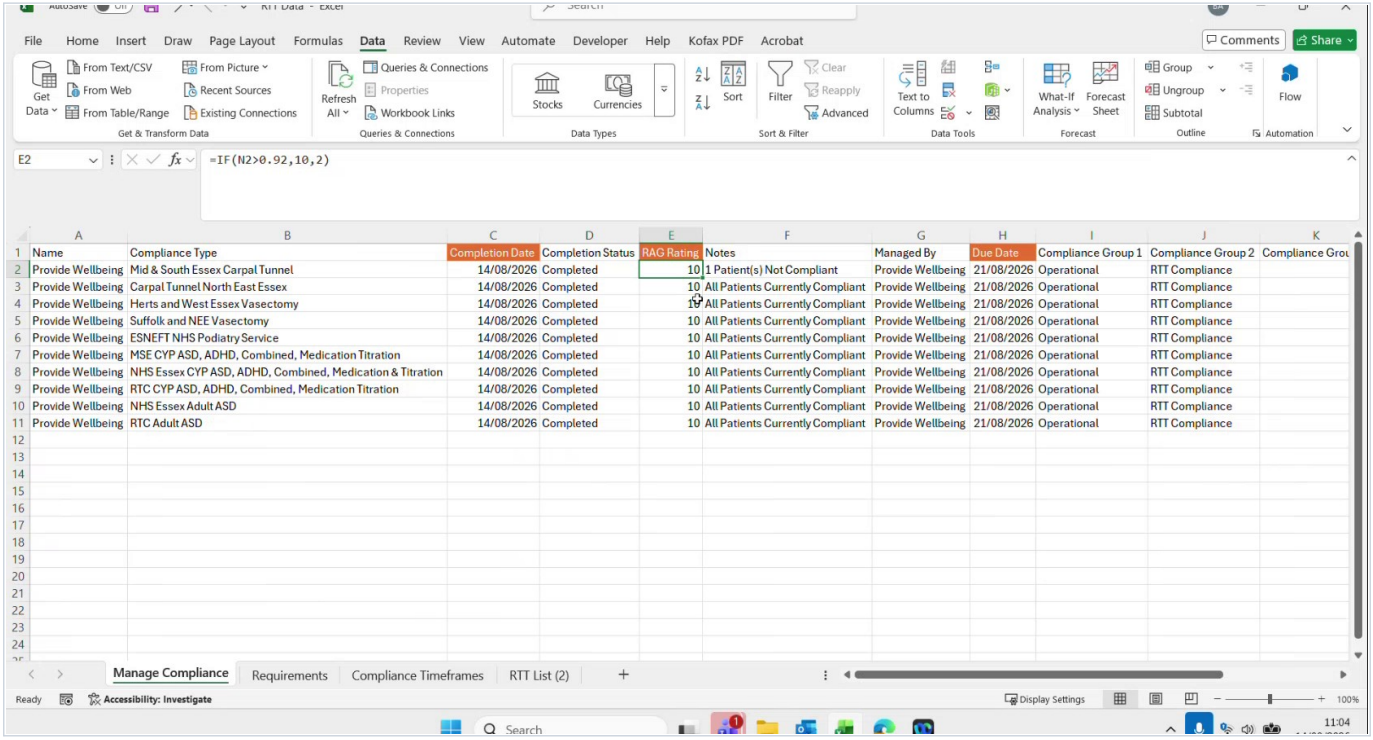



**Figure 18 [00:06:09] — RTT Data.xlsx › Manage Compliance:** Formula bar, E2 RAG Rating =IF(N2>0.92,10,2): 10 = compliant, 2 = not compliant; N2 = 99%, NC Patients = 1

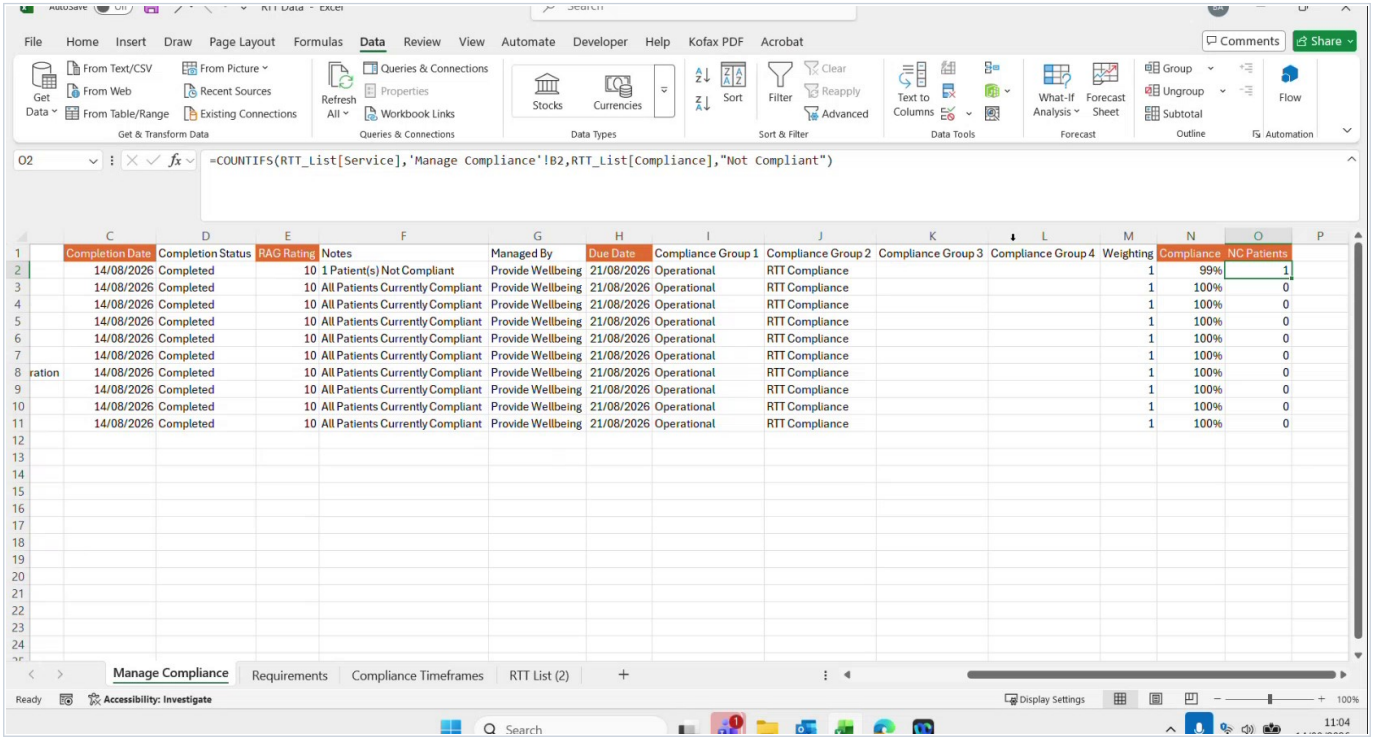



**Figure 19 [00:06:31] — RTT Data.xlsx › Manage Compliance:** Formula bar, O2 NC Patients =COUNTIFS(RTT_List[Service],'Manage Compliance'!B2,RTT_List[Compliance],"Not Compliant")

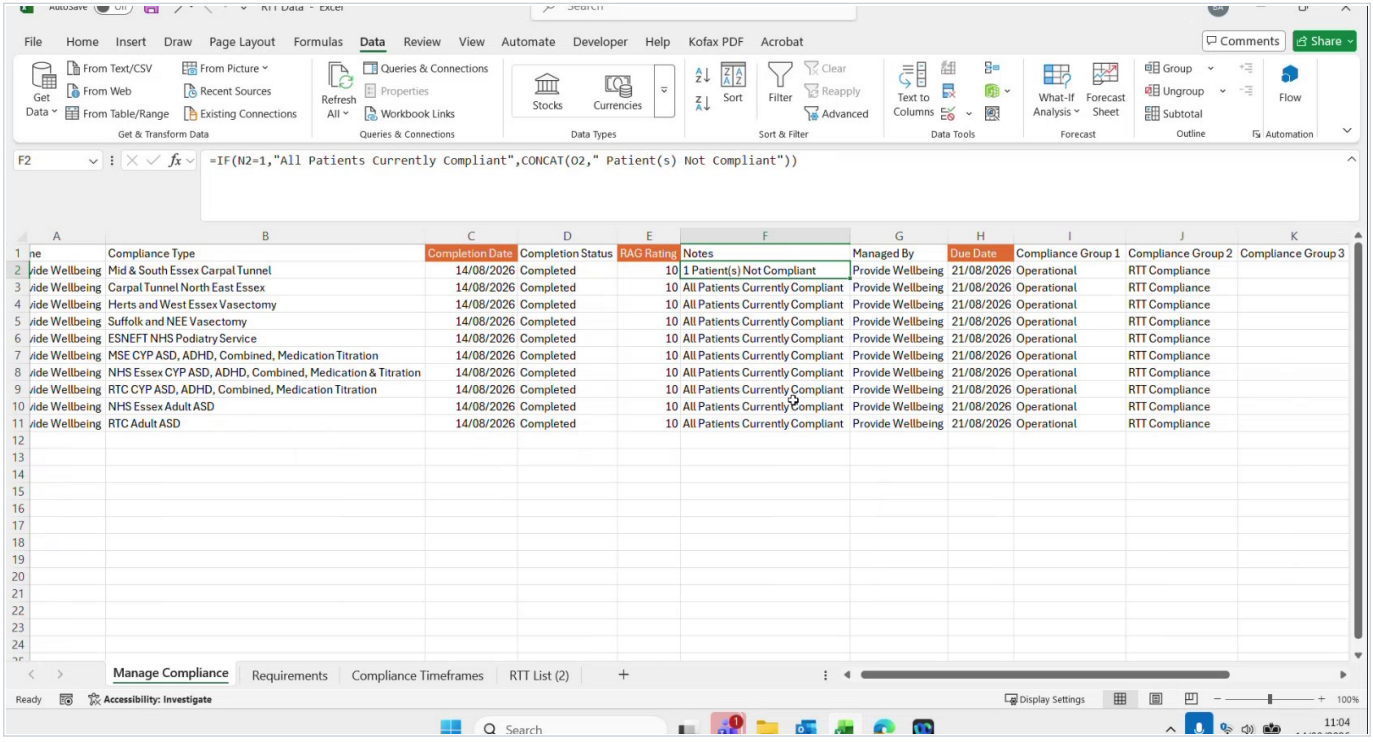



**Figure 20 [00:06:39] — RTT Data.xlsx › Manage Compliance:** Formula bar, F2 Notes =IF(N2=1,"All Patients Currently Compliant",CONCAT(O2," Patient(s) Not Compliant"))

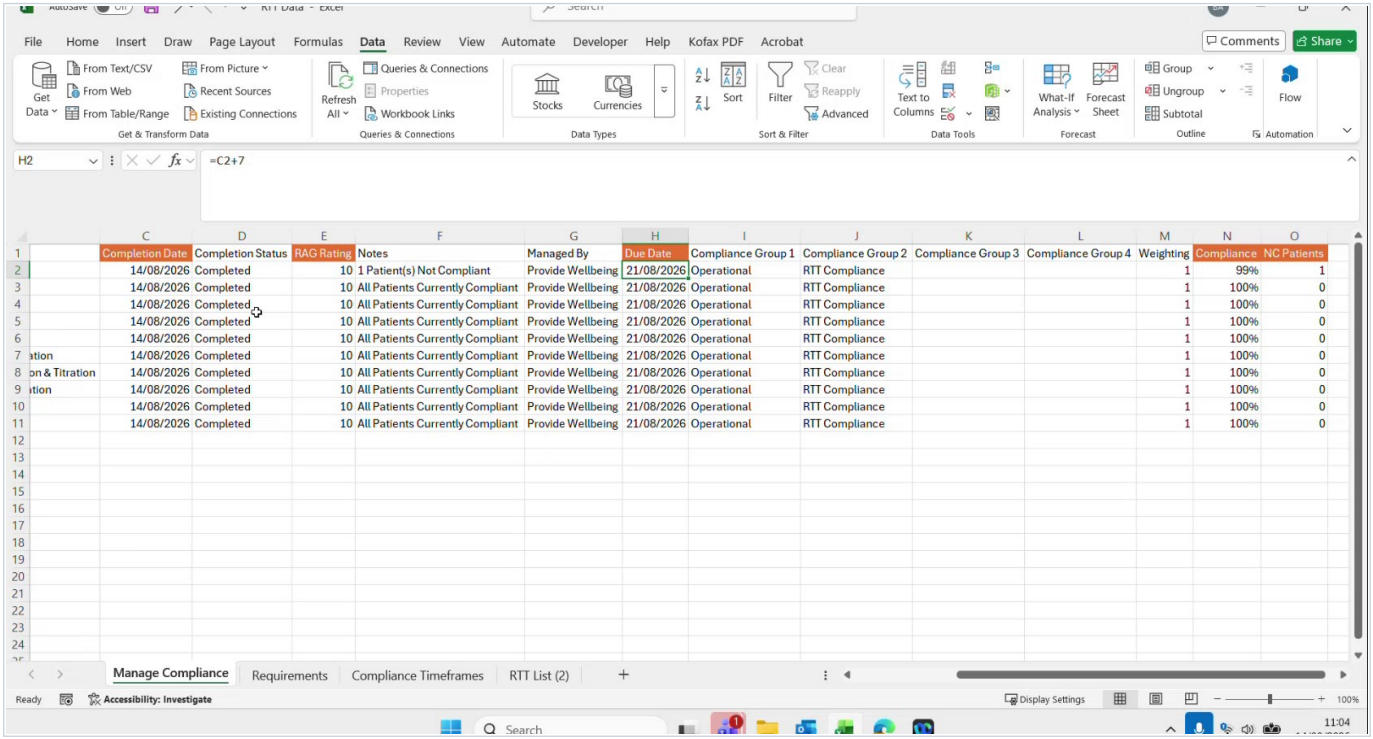



**Figure 21 [00:06:48] — RTT Data.xlsx › Manage Compliance:** Formula bar, H2 Due Date =C2+7 (one week)

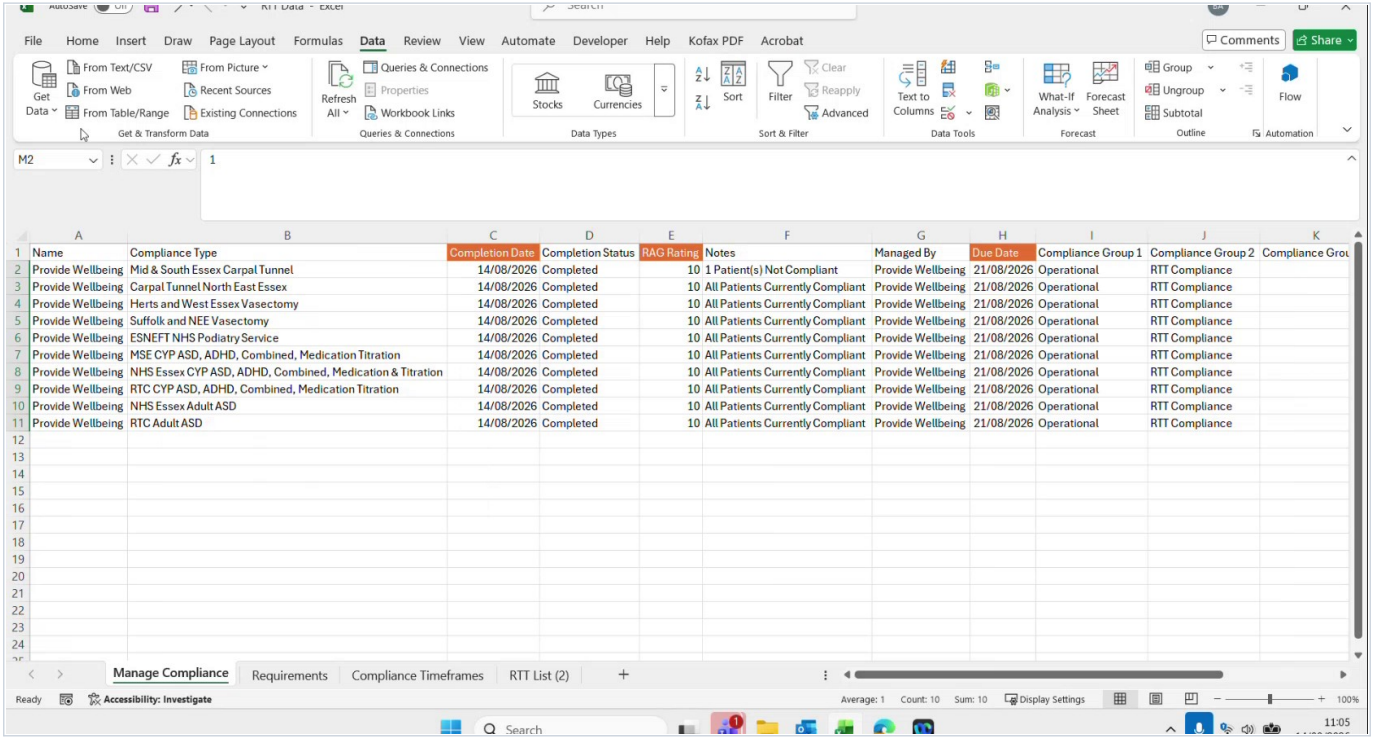



**Figure 22 [00:07:13] — RTT Data.xlsx › Manage Compliance:** M2 Weighting = 1 (“every weighting seems to be one... Only Matt knows”); rows selected for copying

## 2.4 Compliance Data.xlsm

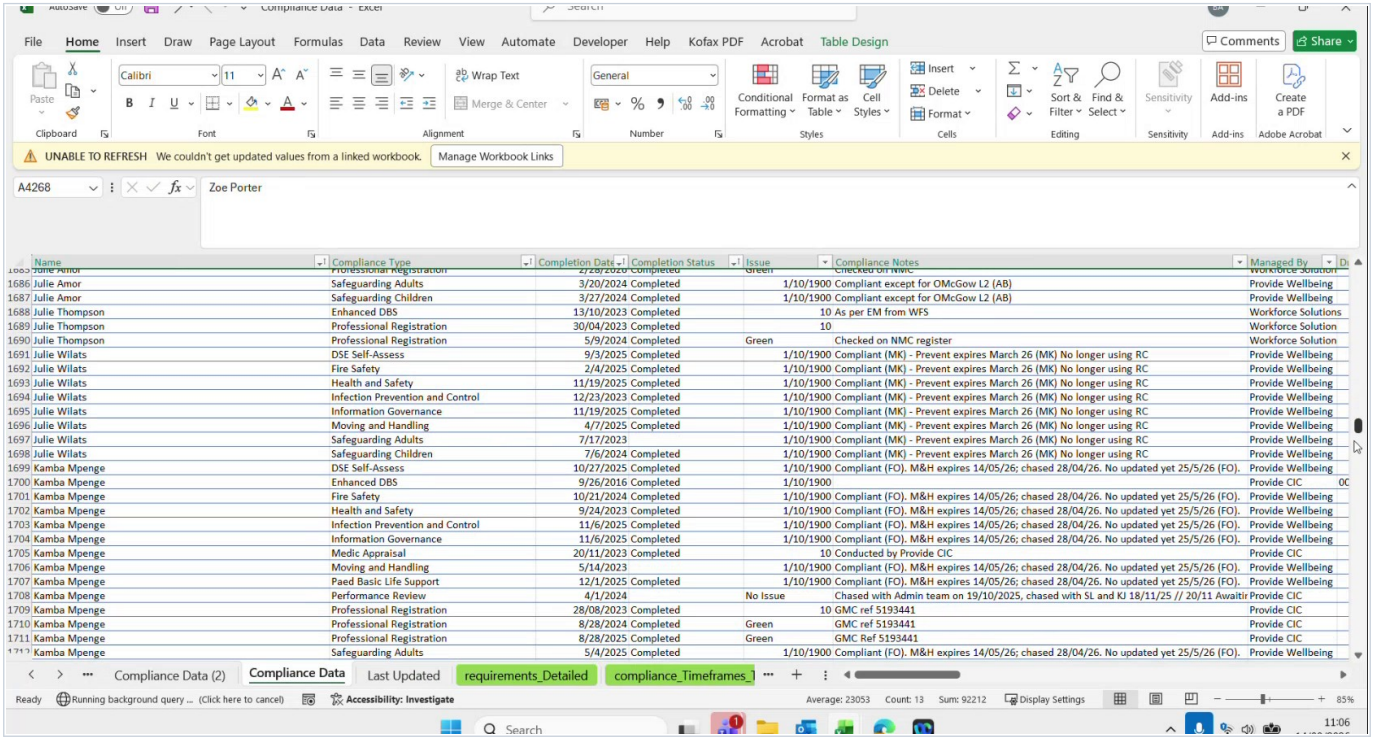



**Figure 24 [00:07:59] — Compliance Data.xlsm › Compliance Data:** Compliance Data (UNABLE TO REFRESH banner) at the last row 4268, where the rows are pasted; paste not performed on screen

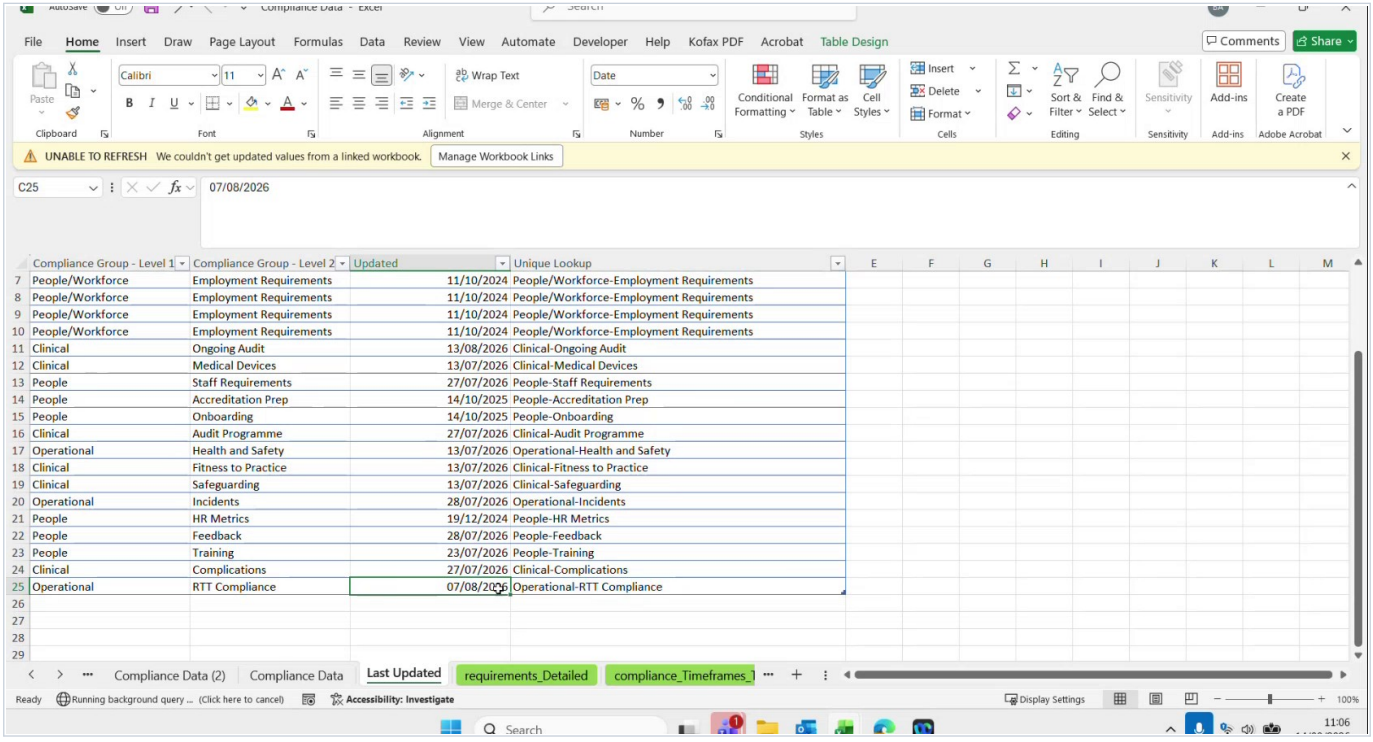



**Figure 25 [00:08:08] — Compliance Data.xlsm › Last Updated:** Last Updated: Operational-RTT Compliance 07/08/2026 (“we’ll fill in our last updated, it should be today”)

# 3. Chronological Step-by-Step Operator Workflow

Eight steps cover the 8:42 recording. Step 6 changed the production workbook; the weekly paste and refresh are described only.

| Step | Window | Action |
|---|---|---|
| 1 | 00:00:00–00:00:40 | Introduce RTT compliance |
| 2 | 00:00:40–00:01:19 | Requirements |
| 3 | 00:01:19–00:02:24 | Compliance Timeframes and thresholds |
| 4 | 00:02:24–00:03:24 | Source: RTT breach report via Power Query |
| 5 | 00:03:24–00:04:07 | Columns of the RTT List |
| 6 | 00:04:07–00:05:41 | **Live edit**: delete Breach Date 2 and rewrite RTT Week |
| 7 | 00:05:41–00:07:10 | Explain the Manage Compliance rows |
| 8 | 00:07:10–00:08:42 | Copy into the data store; last updated; save; refresh twice (described) |

## Step 1 [00:00:00 – 00:00:40]: Introduce RTT compliance

| | |
|---|---|
| **Action performed** | “RTT compliance for the whole of Provide Wellbeing... no need for names, one entity.” Manage Compliance on screen. |
| **Source → target** | — |
| **State mutation** | — |

*Shown above: Figure 1 [00:00:05] — Recording starts: “another video to show you how we update compliance data for RTT… for the whole of Provide W…*

*Shown above: Figure 2 [00:00:15] — Manage Compliance: one row per service (10), completed 14/08/2026, RAG 10, Notes such as “1 Patient(s) Not Com…*

## Step 2 [00:00:40 – 00:01:19]: Requirements

| | |
|---|---|
| **Action performed** | Ten services, required, core, unique lookup, active. |
| **Source → target** | — |
| **State mutation** | — |

*Shown above: Figure 3 [00:01:14] — Requirements: Provide Wellbeing × 10 services (Mid & South Essex Carpal Tunnel, Carpal Tunnel North East Essex…*

## Step 3 [00:01:19 – 00:02:24]: Compliance Timeframes and thresholds

| | |
|---|---|
| **Action performed** | 0.25 months (“every week, four times a month”), Operational › RTT Compliance, lookups “copied from the previous sheet”, thresholds (green 100%, red below 92%). |
| **Source → target** | — |
| **State mutation** | — |

*Shown above: Figure 4 [00:01:25] — Compliance Timeframes: Every x months = 0.25 (“we update this every week, so it’s four times a month”), Core, …*

*Shown above: Figure 5 [00:02:02] — Formula bar, H2 Unique Lookup =CONCAT([@[Compliance Group 1]],"-",[@[Compliance Group 2]],"-",[@[Compliance Gr…*

*Shown above: Figure 6 [00:02:04] — Formula bar, I3 Missing =XLOOKUP(CONCAT(add_Requirements_Name,"-",[@[Compliance Type]]),[Unique Lookup],[Uniqu…*

*Shown above: Figure 7 [00:02:10] — Thresholds J:N = 1 / 0.98 / 0.96 / 0.94 / 0.92: green at 100%, red below 92% (“shades in between”)*

## Step 4 [00:02:24 – 00:03:24]: Source: RTT breach report via Power Query

| | |
|---|---|
| **Action performed** | Explains the weekly breach report built from SystmOne; opens the query editor (RTT List). |
| **Source → target** | SystmOne → breach report → query → RTT_List |
| **State mutation** | Refresh not shown |

*Shown above: Figure 8 [00:02:31] — RTT List (query output RTT_List) with the Service filter menu open: NHS Number (blurred), Breach Date, Service…*

*Shown above: Figure 9 [00:02:56] — Query RTT List: last step =Table.SelectRows(#"Filtered Rows", each [NHS Number] <> 0); source is the weekly RT…*

## Step 5 [00:03:24 – 00:04:07]: Columns of the RTT List

| | |
|---|---|
| **Action performed** | NHS number, breach date, service, RTT week; “What’s column 43? Can’t actually remember.” |
| **Source → target** | — |
| **State mutation** | — |

*Shown above: Figure 10 [00:03:47] — Column43 (e.g. 26.857, 11.857) — “What’s column 43? Can’t actually remember”: a weeks-waited value carried fro…*

## Step 6 [00:04:07 – 00:05:41]: Live edit: delete Breach Date 2 and rewrite RTT Week

| | |
|---|---|
| **Action performed** | “The breach date is artificially calculated... we don’t need a breach date... we’ll delete that.” Column deleted → #REF!; formula retyped with ROUNDUP (19, Not Compliant), then ROUNDDOWN (18, Compliant). “Sorry about these running repairs.” |
| **Source → target** | Formula change in production workbook |
| **State mutation** | First patient: Not Compliant → Compliant; service adjustments for ECC Autism and Norfolk & Waveney ASD removed |

*Shown above: Figure 11 [00:04:01] — BEFORE the live edit: Breach Date 2 =IF([@Service]="ECC Autism in Schools",[@[Breach Date]]+70,IF([@Service]="…*

*Shown above: Figure 12 [00:04:20] — BEFORE: RTT Week F2 =18-(([@[Breach Date 2]]-TODAY())/7); first patient: breach 12/08/2026, RTT Week 18, Not C…*

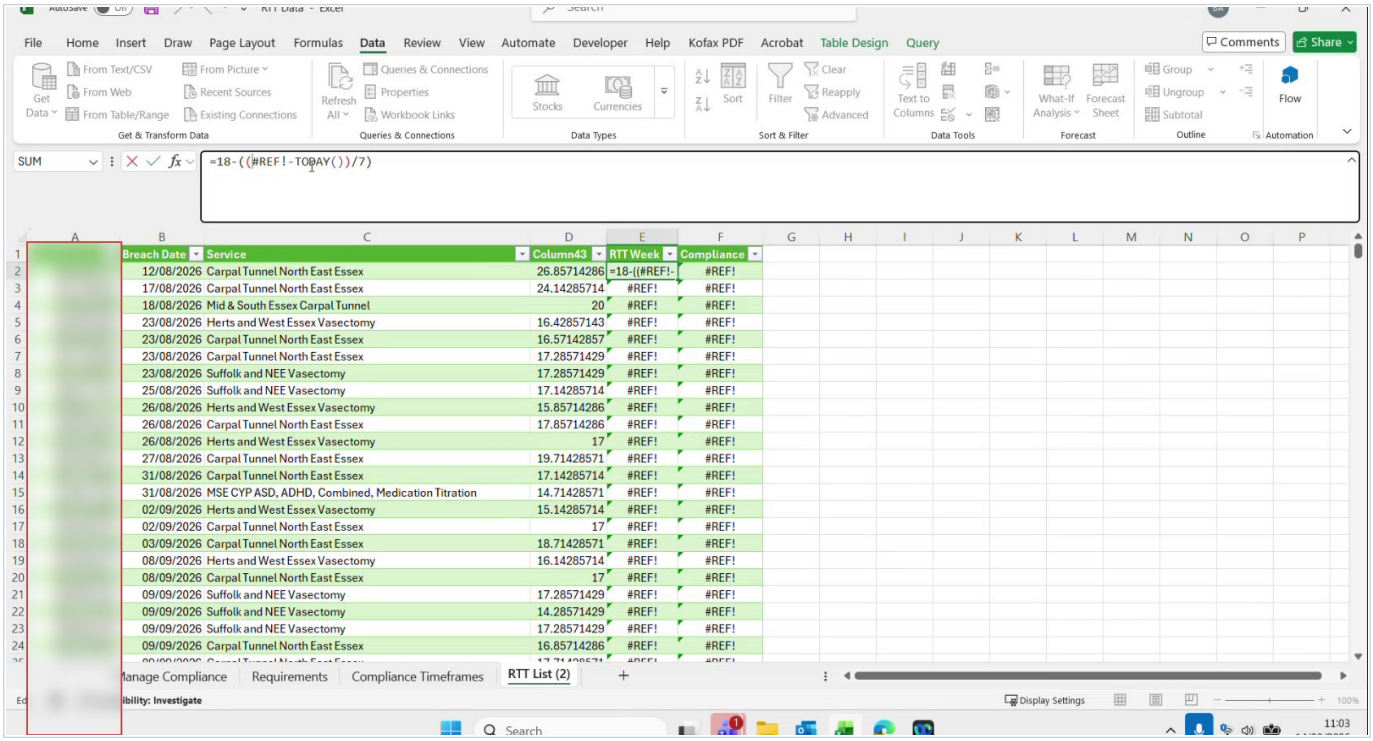



**Figure 13 [00:04:53] — RTT Data.xlsx › RTT List:** Operator deletes Breach Date 2 (“we don’t need a breach date”): RTT Week and Compliance become #REF!, formula =18-((#REF!-TODAY())/7) edited in the cell

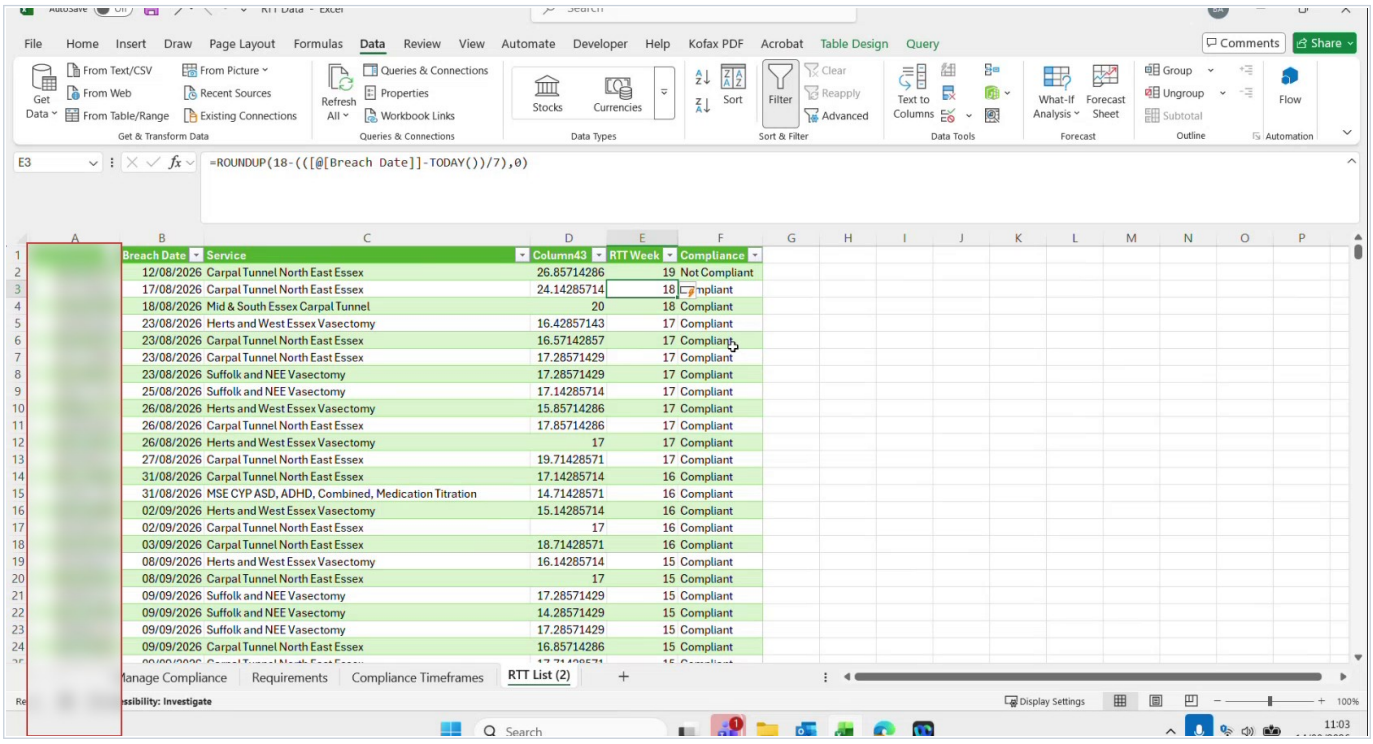



**Figure 14 [00:05:24] — RTT Data.xlsx › RTT List:** Rewritten as =ROUNDUP(18-(([@[Breach Date]]-TODAY())/7),0): first patient now RTT Week 19, Not Compliant

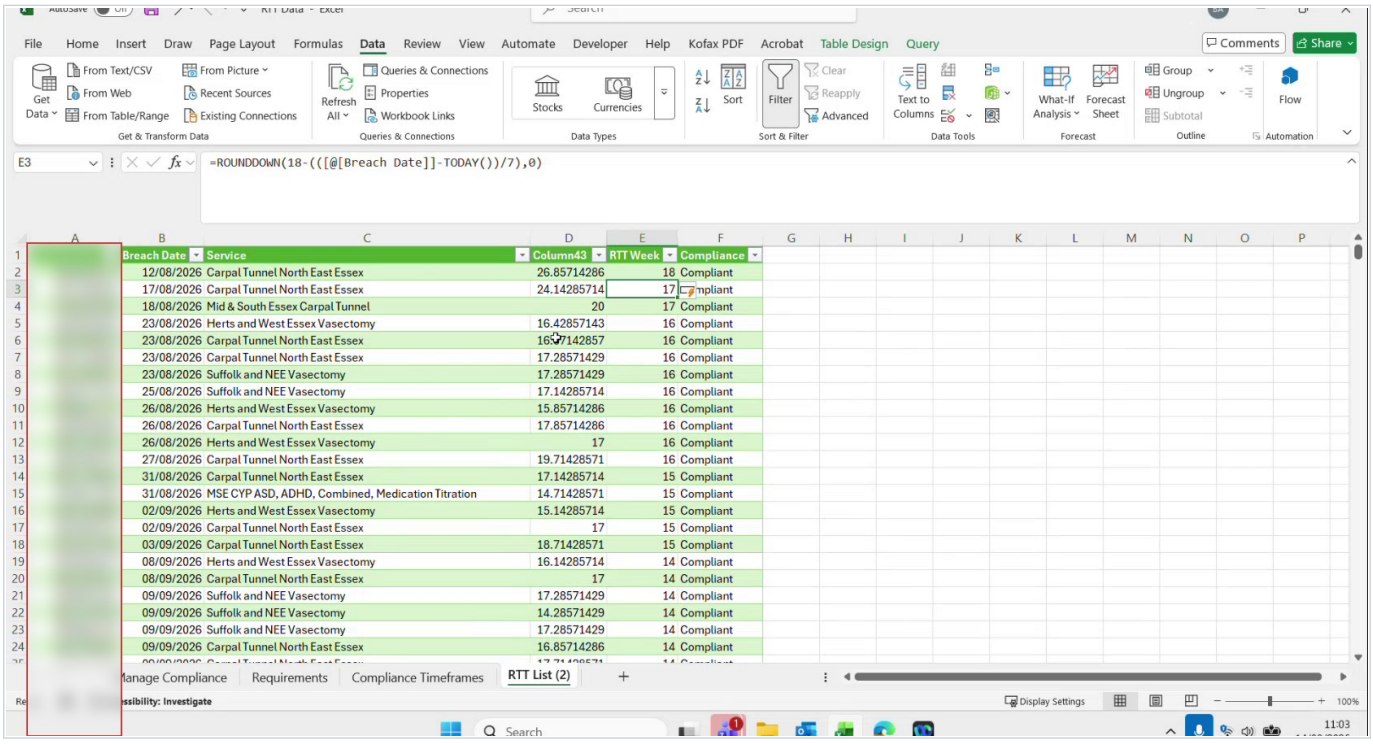



**Figure 15 [00:05:41] — RTT Data.xlsx › RTT List:** AFTER: changed to =ROUNDDOWN(18-(([@[Breach Date]]-TODAY())/7),0) (“we’ll have it to round down, I think”): the same patient, two days past the breach date, now shows 18 and Compliant

## Step 7 [00:05:41 – 00:07:10]: Explain the Manage Compliance rows

| | |
|---|---|
| **Action performed** | Completion date today, status Completed, RAG (>92% → 10 else 2), notes from the NC count, due date +7, Operational › RTT Compliance, weighting 1 (“Only Matt knows”). |
| **Source → target** | RTT_List → Manage Compliance |
| **State mutation** | — |

*Shown above: Figure 16 [00:05:46] — Formula bar, N2 Compliance %: COUNTIFS(… "Compliant")/COUNTIF(…) per service from RTT_List*

*Shown above: Figure 17 [00:05:52] — Formula bar, C4 Completion Date =TODAY()*

*Shown above: Figure 18 [00:06:09] — Formula bar, E2 RAG Rating =IF(N2>0.92,10,2): 10 = compliant, 2 = not compliant; N2 = 99%, NC Patients = 1*

*Shown above: Figure 19 [00:06:31] — Formula bar, O2 NC Patients =COUNTIFS(RTT_List[Service],'Manage Compliance'!B2,RTT_List[Compliance],"Not Compl…*

*Shown above: Figure 20 [00:06:39] — Formula bar, F2 Notes =IF(N2=1,"All Patients Currently Compliant",CONCAT(O2," Patient(s) Not Compliant"))*

*Shown above: Figure 21 [00:06:48] — Formula bar, H2 Due Date =C2+7 (one week)*

*Shown above: Figure 22 [00:07:13] — M2 Weighting = 1 (“every weighting seems to be one… Only Matt knows”); rows selected for copying*

## Step 8 [00:07:10 – 00:08:42]: Copy into the data store; last updated; save; refresh twice (described)

| | |
|---|---|
| **Action performed** | Selects the rows, reopens Compliance Data (“accidentally closed”), shows the bottom row and Last Updated (Operational-RTT Compliance 07/08/2026). Paste and edit not performed. |
| **Source → target** | Manage Compliance → Compliance Data |
| **State mutation** | Not performed |

*Shown above: Figure 23 [00:07:25] — Reopening Compliance Data (“which I must have accidentally closed”): folders Archive, Automated, Dashboard, Da…*

*Shown above: Figure 24 [00:07:59] — Compliance Data (UNABLE TO REFRESH banner) at the last row 4268, where the rows are pasted; paste not performe…*

*Shown above: Figure 25 [00:08:08] — Last Updated: Operational-RTT Compliance 07/08/2026 (“we’ll fill in our last updated, it should be today”)*

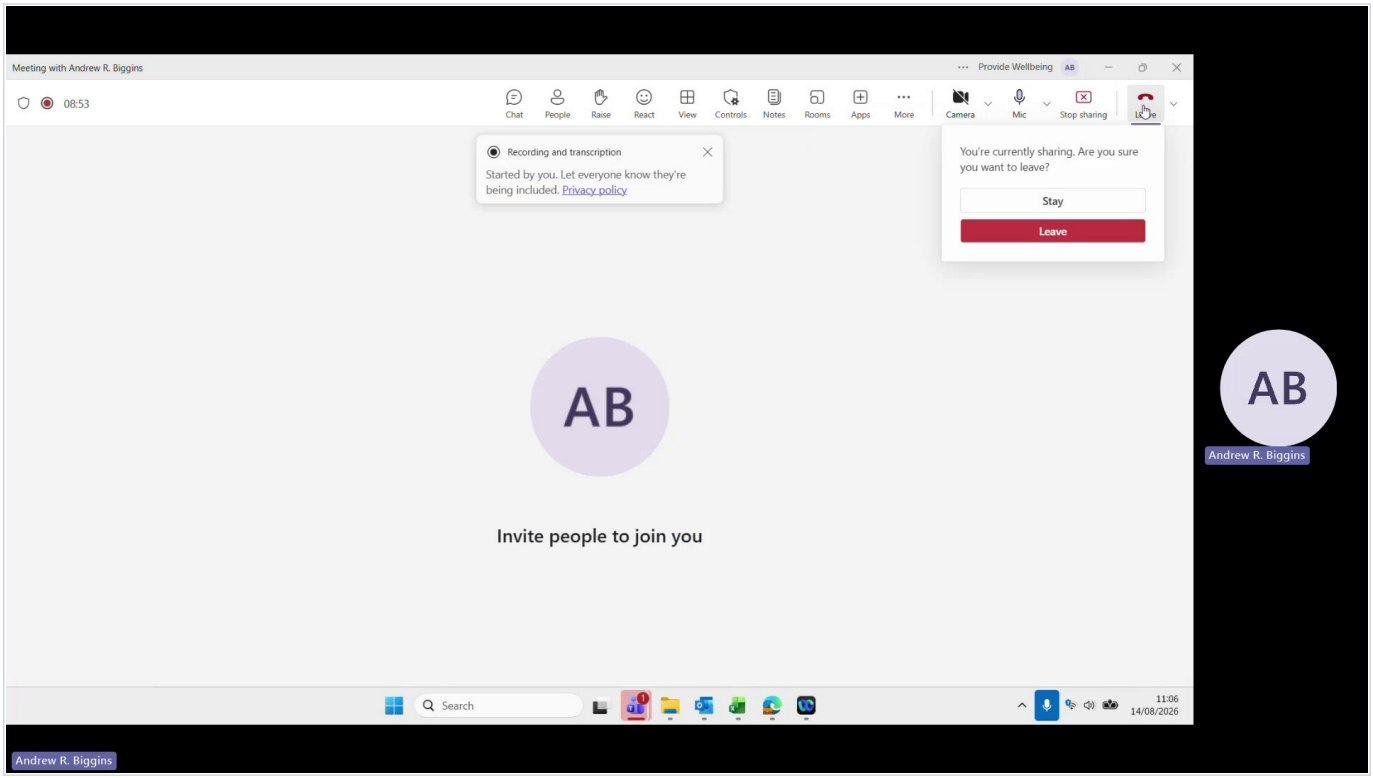



**Figure 26 [00:08:40] — Teams recording › (end):** Screen share stopped; recording ends at 08:42

# 4. Data Quality Checks, Manual Exceptions & Edge Cases

## 4.1 Business rules

| ID | Rule | Evidence |
|---|---|---|
| BR-01 | RTT standard: patients should start treatment within 18 weeks of referral; RTT Week = 18 − weeks remaining to the breach date | Figure 12 [00:04:20] |
| BR-02 | Some services have a different breach date (ECC Autism in Schools +70 days, Norfolk and Waveney ASD +42 days) — removed during the recording | Figure 11 [00:04:01] |
| BR-03 | Service compliance % = Compliant patients ÷ all patients on the list for the service | Figure 16 [00:05:46] |
| BR-04 | RAG 10 if compliance % > 92%, else 2 | Figure 18 [00:06:09] |
| BR-05 | Weekly cadence; due date = today + 7 | Figure 21 [00:06:48] |

## 4.2 Manual steps

| ID | Manual step | Risk |
|---|---|---|
| EX-01 | Weekly SystmOne download and breach report built by the operational team | Upstream errors flow straight in |
| EX-02 | Rows copied to the bottom of Compliance Data each week | Duplicates or missed weeks |
| EX-03 | Last Updated edited by hand (still 07/08/2026 on 14/08) | Stale timestamp |
| EX-04 | Formulas edited ad hoc in the production file | Unreviewed rule changes (see SV-01) |

## 4.3 Spoken vs visual discrepancies and defects

| ID | Said / intended | Seen | Impact |
|---|---|---|---|
| SV-01 | “The breach date is artificially calculated… we don’t need it” (00:04:17–00:04:47) | Breach Date 2 held service-specific +70 / +42 day rules; deleting it removed them | ECC Autism and Norfolk & Waveney ASD patients now measured against the wrong breach date |
| SV-02 | “We’ll have it to round down, I think” (00:05:28) | ROUNDDOWN turns 18.29 weeks into 18; the patient 2 days past breach becomes Compliant | Breaches up to 6 days late are hidden; compliance overstated |
| SV-03 | “If it’s less than 92 it will be red” (00:02:17) | RAG =IF(N2>0.92,10,2): exactly 92% is red | Boundary disagrees |
| SV-04 | Column 43 (“can’t actually remember”) | Decimal weeks value carried from the report | Unused or misunderstood data |
| SV-05 | Weighting “always one… only Matt knows” | Constant 1 | Meaning unknown |
| FD-01 | Stable weekly result | TODAY() in RTT Week and Completion Date | Re-opening on another day changes results |
| FD-02 | Compliance % per service | Division by COUNTIF; a service with no patients gives #DIV/0! | Error rows in the store |
| FD-03 | Usual constants | Missing TRUE, trailing-dash lookup, UNABLE TO REFRESH | As in specs 01–05 |

## 4.4 Hidden failure modes

| ID | Failure mode | Severity |
|---|---|---|
| FM-01 | **Uncontrolled rule changes** in a live file, with no review or version history | High |
| FM-02 | **Lost service-specific breach rules** | High |
| FM-03 | **Rounding hides breaches** | High |
| FM-04 | **Patient identifiers** (NHS numbers) in a reporting workbook | High (IG) |
| FM-05 | **Dependency on a manually built weekly report** | Medium |
| FM-06 | **Non-reproducible weekly results** (TODAY()) | Medium |

# 5. Target Modern System Architecture Blueprint

RTT runs on the shared platform (spec 01 §5). Patient-level data stays in a restricted zone; only service-level counts are published.

## 5.1 Ingestion

- Scheduled SystmOne waiting-list extract (or the breach report file) with a header contract; NHS number validated (checksum) and kept restricted.
- Each weekly load is a snapshot with an as-of date, so past weeks can be recomputed.

## 5.2 Data model

| Table | Key columns |
|---|---|
| clin.rtt_pathway_snapshot | as_of_date, pathway_id, nhs_number (restricted), service_id, clock_start or breach_date |
| ref.rtt_service_rule (versioned) | service_id, breach_offset_days (e.g. +70, +42), standard_weeks (18), valid_from/to, approved_by |
| ref.rtt_threshold (versioned) | rag rule (> or ≥ 92%), dashboard bands |
| mart.rtt_service_week | as_of_date, service_id, patients, compliant, not_compliant, pct, rag |

## 5.3 Rules as SQL

```
SELECT s.as_of_date, s.service_id,
       count(*) AS patients,
       count(*) FILTER (WHERE s.as_of_date <= s.breach_date + r.breach_offset_days) AS compliant
FROM clin.rtt_pathway_snapshot s
JOIN ref.rtt_service_rule r ON r.service_id = s.service_id
 AND s.as_of_date BETWEEN r.valid_from AND coalesce(r.valid_to,'infinity')
GROUP BY 1,2;
-- breach compared on dates, so no rounding can hide a late patient
```

- Rule changes (offsets, thresholds, rounding) are versioned reference data with an approver, never edits to a live formula (fixes FM-01, FM-02).
- Compliance is decided on dates, not rounded weeks (fixes FM-03); a service with no patients publishes “no patients” rather than #DIV/0!.
- Weekly publish writes ten rows with the as-of date inside the shared atomic publish; Last Updated is automatic.

# 6. Gap & Ambiguity Register

| ID | Timestamp | Gap / ambiguity | Ask |
|---|---|---|---|
| G-01 | 00:04:17–00:05:34 | Was the live change (Breach Date 2 removed, ROUNDDOWN) kept and saved? | Confirm and, if needed, restore the previous version |
| G-02 | 00:03:57 | Why ECC Autism in Schools (+70) and Norfolk and Waveney ASD (+42) had adjusted breach dates; are these services still in scope? | Clinical/operational lead to confirm |
| G-03 | 00:05:04–00:05:28 | Round up, round down or compare dates? | Business owner to decide |
| G-04 | 00:03:43 | Meaning of Column43 | Explain or drop |
| G-05 | 00:02:26 | Name, location and build steps of the RTT breach report | Share the file and process |
| G-06 | 00:06:07 | RAG boundary at exactly 92% | Confirm |
| G-07 | 00:07:04 | Meaning of Weighting = 1 | Define |
| G-08 | whole video | Compliance column formula on RTT List (not shown) | Share the workbook |
| G-09 | file name | Original title and date of the recording (MEDDAD~1) | Confirm |

# 7. What Is Required to Build the Modern System

RTT-specific requirements with reasons; shared platform items are in spec 01 §7. Client checklist: **RTT_PreBuild_Client_Requirements.pdf**.

## 7.1 Functional requirements

| ID | Requirement (priority) | Why it is required |
|---|---|---|
| FR-R01 | Scheduled SystmOne RTT extract with weekly snapshots (M) | Manual download and report build today (EX-01). |
| FR-R02 | Versioned per-service breach rules (M) | The +70/+42 rules were deleted live (Figure 11 [00:04:01]). |
| FR-R03 | Date-based breach test, no rounding (M) | ROUNDDOWN hid a breach (Figure 15 [00:05:41]). |
| FR-R04 | Generated weekly compliance rows with counts (M) | Replaces Manage Compliance (Figure 2 [00:00:15]). |
| FR-R05 | Change control for rule changes (approval, history) (M) | Live edits in the production file (FM-01). |
| FR-R06 | Patient drill-down for operational teams, restricted (S) | The breach report is also used to prevent breaches. |

## 7.2 Non-functional and testing

| ID | Requirement | Why it is required |
|---|---|---|
| NFR-R01 | NHS numbers restricted; dashboard sees service counts only | Patient identifiers in the workbook (FM-04). |
| NFR-R02 | Reproducible weekly snapshots | TODAY() makes results day-dependent (FD-01). |
| T-R01 | Golden case: patient with breach 12/08/2026 on 14/08/2026 must be Not Compliant | The exact case the live edit flipped. |
| T-R02 | Service offsets: ECC Autism +70, Norfolk and Waveney ASD +42 (if confirmed) | BR-02. |
| T-R03 | Boundary at 92% and empty services | SV-03, FD-02. |

# Appendix A. Figure Index

| Fig | Time | File › Tab | Shows |
|---|---|---|---|
| 1 | 00:00:05 | Teams recording › (camera off) | Recording starts: “another video to show you how we update compliance data for RTT… for the whole of Provide Wellbeing” |
| 2 | 00:00:15 | RTT Data.xlsx › Manage Compliance | Manage Compliance: one row per service (10), completed 14/08/2026, RAG 10, Notes such as “1 Patient(s) Not Compliant”, due 21/08/2026, Operational › R… |
| 3 | 00:01:14 | RTT Data.xlsx › Requirements | Requirements: Provide Wellbeing × 10 services (Mid & South Essex Carpal Tunnel, Carpal Tunnel North East Essex, Herts and West Essex Vasectomy, Su… |
| 4 | 00:01:25 | RTT Data.xlsx › Compliance Timeframes | Compliance Timeframes: Every x months = 0.25 (“we update this every week, so it’s four times a month”), Core, Operational › RTT Compliance |
| 5 | 00:02:02 | RTT Data.xlsx › Compliance Timeframes | Formula bar, H2 Unique Lookup =CONCAT([@[Compliance Group 1]],"-",[@[Compliance Group 2]],"-",[@[Compliance Group 3]]) → “Operatio… |
| 6 | 00:02:04 | RTT Data.xlsx › Compliance Timeframes | Formula bar, I3 Missing =XLOOKUP(CONCAT(add_Requirements_Name,"-",[@[Compliance Type]]),[Unique Lookup],[Unique Lookup],"")="… |
| 7 | 00:02:10 | RTT Data.xlsx › Compliance Timeframes | Thresholds J:N = 1 / 0.98 / 0.96 / 0.94 / 0.92: green at 100%, red below 92% (“shades in between”) |
| 8 | 00:02:31 | RTT Data.xlsx › RTT List | RTT List (query output RTT_List) with the Service filter menu open: NHS Number (blurred), Breach Date, Service, Column43, Breach Date 2, RTT Week, Com… |
| 9 | 00:02:56 | RTT Data.xlsx › Power Query editor | Query RTT List: last step =Table.SelectRows(#"Filtered Rows", each [NHS Number] <> 0); source is the weekly RTT breach report built fr… |
| 10 | 00:03:47 | RTT Data.xlsx › RTT List | Column43 (e.g. 26.857, 11.857) — “What’s column 43? Can’t actually remember”: a weeks-waited value carried from the breach report [Inferred] |
| 11 | 00:04:01 | RTT Data.xlsx › RTT List | BEFORE the live edit: Breach Date 2 =IF([@Service]="ECC Autism in Schools",[@[Breach Date]]+70,IF([@Service]="Norfolk and Waveney - ASD… |
| 12 | 00:04:20 | RTT Data.xlsx › RTT List | BEFORE: RTT Week F2 =18-(([@[Breach Date 2]]-TODAY())/7); first patient: breach 12/08/2026, RTT Week 18, Not Compliant |
| 13 | 00:04:53 | RTT Data.xlsx › RTT List | Operator deletes Breach Date 2 (“we don’t need a breach date”): RTT Week and Compliance become #REF!, formula =18-((#REF!-TODAY())/7) edited in the ce… |
| 14 | 00:05:24 | RTT Data.xlsx › RTT List | Rewritten as =ROUNDUP(18-(([@[Breach Date]]-TODAY())/7),0): first patient now RTT Week 19, Not Compliant |
| 15 | 00:05:41 | RTT Data.xlsx › RTT List | AFTER: changed to =ROUNDDOWN(18-(([@[Breach Date]]-TODAY())/7),0) (“we’ll have it to round down, I think”): the same patient, two days past the breach… |
| 16 | 00:05:46 | RTT Data.xlsx › Manage Compliance | Formula bar, N2 Compliance %: COUNTIFS(… "Compliant")/COUNTIF(…) per service from RTT_List |
| 17 | 00:05:52 | RTT Data.xlsx › Manage Compliance | Formula bar, C4 Completion Date =TODAY() |
| 18 | 00:06:09 | RTT Data.xlsx › Manage Compliance | Formula bar, E2 RAG Rating =IF(N2>0.92,10,2): 10 = compliant, 2 = not compliant; N2 = 99%, NC Patients = 1 |
| 19 | 00:06:31 | RTT Data.xlsx › Manage Compliance | Formula bar, O2 NC Patients =COUNTIFS(RTT_List[Service],'Manage Compliance'!B2,RTT_List[Compliance],"Not Compliant") |
| 20 | 00:06:39 | RTT Data.xlsx › Manage Compliance | Formula bar, F2 Notes =IF(N2=1,"All Patients Currently Compliant",CONCAT(O2," Patient(s) Not Compliant")) |
| 21 | 00:06:48 | RTT Data.xlsx › Manage Compliance | Formula bar, H2 Due Date =C2+7 (one week) |
| 22 | 00:07:13 | RTT Data.xlsx › Manage Compliance | M2 Weighting = 1 (“every weighting seems to be one… Only Matt knows”); rows selected for copying |
| 23 | 00:07:25 | File Explorer › Z:\Performance Dashboards and Data | Reopening Compliance Data (“which I must have accidentally closed”): folders Archive, Automated, Dashboard, Data Store, User input, VBA |
| 24 | 00:07:59 | Compliance Data.xlsm › Compliance Data | Compliance Data (UNABLE TO REFRESH banner) at the last row 4268, where the rows are pasted; paste not performed on screen |
| 25 | 00:08:08 | Compliance Data.xlsm › Last Updated | Last Updated: Operational-RTT Compliance 07/08/2026 (“we’ll fill in our last updated, it should be today”) |
| 26 | 00:08:40 | Teams recording › (end) | Screen share stopped; recording ends at 08:42 |

# Appendix B. Key timestamped utterances

| Time | Utterance (lightly cleaned) |
|---|---|
| 00:00:03 | This is another video to show you how we update compliance data for RTT. |
| 00:00:26 | As with incidents management, there’s no need for names… one entity, Provide Wellbeing. |
| 00:01:36 | We update this every week, so it’s four times a month. |
| 00:02:04 | I just copied that from the previous sheet. |

| Time | Utterance (lightly cleaned) |
|---|---|
| 00:02:15 | It will be green if it’s 100%. If it’s less than 92, it will be red. |
| 00:02:26 | It pulls in data from another file, which is our RTT breach report… downloaded from SystmOne. |
| 00:03:43 | What’s column 43? Can’t actually remember. |
| 00:04:17 | The breach date, sorry, is artificially calculated. |
| 00:04:42 | We don’t need a breach date… so we’ll delete that. |
| 00:04:50 | Sorry about these running repairs. |
| 00:05:04 | We probably want that to round up… or do we want it to round down? We’ll have it to round down, I think. |
| 00:06:07 | If it’s less than 92% it’s going to give us a RAG rating of two, which is not compliant. |
| 00:06:47 | Due date is the completion date plus seven days. |
| 00:07:01 | Every weighting seems to be one… Only Matt knows. |
| 00:07:54 | We’ll copy it and paste it into the bottom of our compliance data. |
| 00:08:14 | We would save, refresh the tables twice. |